

A
Project Report
on
**MedTalk: Explanation-Driven Adaptive Questioning for
Responsible Interpretable AI Across Regulated Sectors**

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We further declare that the work reported in this report has not been used as the basis for any other examination, in this or any other university. This report is submitted as partial fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering at KIET Group of Institutions, Ghaziabad, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow.

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ABSTRACT

Diabetes mellitus is one of the world's most pressing 21st-Century health challenges. 537 million in 2020 with predictions of 783 million by 2045. The cost exceeds 1 trillion USD per year. Type 2 diabetes is mostly preventable by early diagnosis but current diagnostic methods are often reactive and non-transparent. Artificial Intelligence (AI) can provide predictive insights, but the current black-box approach of AI algorithms undermines trust within the clinical community and excludes patients from the management of their own health.

The growing use of ML Models in decision making creates a significant hurdle in establishing trust, and regulatory compliance. Current conversational explainable AI (XAI) systems (such as TalkToModel - TTM), suffer from Large Language Model (LLM) limitations, such as hallucinations and poor calibration of probabilities. In this study we propose MedTalk, a four-layered deterministic approach that overcomes stochastic inference with explainable Random Forest classifiers and a novel Explanation-Driven Adaptive Questioning (EDAQ) algorithm. The EDAQ algorithm proactively guides the clinician's questioning through the measurement of Feature Uncertainty and Significance, enabling the detection of the next follow-up questions that are aimed at clarifying the Risk Assessment.

The research employs the Pima Indians Diabetes Dataset (PIDD) to train and comprehensively optimise a Random Forest classifier with Synthetic Minority Over-sampling Technique (SMOTE) class balancing and RandomizedSearchCV hyperparameter tuning. Experimental analysis on the UCI Pima Indian Diabetes dataset showed MedTalk had an accuracy of 89.4% (95.45% with the SMOTE-optimized version), Brier score of 0.142 (TTM: 0.189), AUC-ROC of 0.991, precision of 94.02%, recall of 95.45%, and F1-score of 94.72%. A user study with 32 clinicians showed a 21% higher comprehension (94.2% vs. 73.1%), higher trust in the tool (6.2/7 vs. 4.1/7 for TTM), and 53% lower time for task completion (134.7 to 63.2 seconds). Ablation study showed EDAQ offers a 12.9% improvement in comprehension, the hybrid selection using SHAP and LIME improves model calibration (Brier 0.142 vs. 0.149), and fairness guardrails avoid 7.3% of discriminatory questions in financial domain tests.

The design of MedTalk demonstrates that the trade-off between the level of model predictability and interpretability is not a hard-and-fast rule in machine learning. Not only does MedTalk allow us to meet the high safety standards demanded for the use in software as medical device (SaMD), it also allows us to reach the level of transparency demanded in patient-centred and ethical medical practice. MedTalk's layer-based architecture with inter-domain configuration without model retraining allows it to be used for healthcare, finance, insurance or judiciary use cases.

Keywords: Explainable AI, Random Forest, SHAP, LIME, Diabetes Prediction, Clinical Decision Support, Responsible AI, EDAQ, TalkToModel, Patient Engagement, Interpretability Gap, Regulatory Compliance, Fairness in AI, Brier Score, AUC-ROC.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-Agnostic Explanations
EDAQ	Explanation-Driven Adaptive Questioning
TTM	TalkToModel
LLM	Large Language Model
RF	Random Forest
PIDD	Pima Indians Diabetes Dataset
SMOTE	Synthetic Minority Over-sampling Technique
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
NLG	Natural Language Generation
SaMD	Software as a Medical Device
FDA	Food and Drug Administration
GDPR	General Data Protection Regulation
UCI	University of California, Irvine
BMI	Body Mass Index
EHR	Electronic Health Record
CDSS	Clinical Decision Support System
NLP	Natural Language Processing
MVP	Minimum Viable Product
USP	Unique Selling Proposition
MDI	Mean Decrease in Impurity
MLLC	Machine Learning Life Cycle
HbA1c	Glycated Haemoglobin
COMPAS	Correctional Offender Management Profiling for Alternative Sanctions
RBF	Radial Basis Function

DPF	Diabetes Pedigree Function
RMSE	Root Mean Squared Error
CAGR	Compound Annual Growth Rate
IDF	International Diabetes Federation
SVM	Support Vector Machine
API	Application Programming Interface
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

CHAPTER 1: INTRODUCTION

1.1 Introduction

The integration of Machine Learning (ML) and Health Care is resulting in unprecedented diagnostic performance. Machine Learning models are routinely outperforming or matching doctors in the domain of diagnosis for well circumscribed activities such as Radiology Image Analysis, Genomic Analysis and Mortality Risk Stratification in Intensive Care Units. But this new technology also brings with it a significant challenge: The most successful AI models are usually the least interpretable; therefore they pose an unbridgeable gap to the acceptance by Doctors of Artificial Intelligence in High Stakes Medical Decision Making Environments.

This project will develop MedTalk, an explanatory, deterministic and holistic AI system to address the "interpretability gap" or the difference between powerful algorithms and trusted practitioners. MedTalk is the convergence of three fields of study: machine learning (ML), explainable AI (XAI) and human computer interaction (HCI). Therefore, MedTalk is a holistic approach to the problem of the interpretability gap in regulated settings. Whereas typical ML projects end with a numeric value, MedTalk provides on-the-fly, patient-specific explanations delivered via a dialogue interface facilitated by the novel EDAQ explanation-driven adaptive questioning algorithm.

The name of the MedTalk System refers to the two-fold nature of it as a Medically-Oriented Prediction Engine (which is built on traditional notions of Deterministic Machine Learning) on one hand and to a Natural Language Dialogue Interface for Clinical Practitioners and Lay patients to communicate with the Systems' Reasoning on the other. The combined emphasis on the Technical Competency of MedTalk and its Communicative Competency is the Design Philosophy of this Project. The MedTalk Thesis is that the Interpretability Gap does not have to be an Inevitable Consequence of Technology, but rather it has been the result of Design Failure which Can be Systematically Addressed by Principled Architectural Decisions.

1.2 The Global Diabetes Crisis

Type II Diabetes Mellitus is one of the biggest health issues today, with Diabetes Mellitus being a chronic metabolic disorder, resulting from abnormally high levels of glucose in the blood leading to significant damage to the blood vessels, heart, eyes, kidneys and nerves. There are different types of Diabetes Mellitus. Type I Diabetes Mellitus is an Autoimmune Disorder where Insulin from the Pancreas is either significantly reduced or absent. Type II Diabetes Mellitus is a Metabolic Disorder that results when there is Insulin Resistance or when the Body cannot utilize Insulin properly. Gestational Diabetes Mellitus develops during Pregnancy and can result in developing Type II Diabetes Mellitus after pregnancy. Globally, Type II Diabetes Mellitus represents approximately 90% of all Diagnosed Cases.

The IDF Diabetes Atlas indicates that about 537 million people worldwide lived with diabetes in the adult population aged 20-79 in 2021. The number represents a diabetic prevalence of 10.5% or almost one tenth of the world's adult population aged 20-79 years old. Projections for the future indicate that if current trends continue, there will be over 643 million adults diagnosed as diabetic by 2030, and over 783 million diabetics by 2045. The speed of this epidemiological transition is unpredictable, the number of people living with diabetes has doubled in just two decades.

The COVID-19 pandemic has caused many problems beyond just an epidemic issue; it has created a large-scale economic problem. According to the IDF, the cost of diabetes to worldwide health expenditures was at least \$966 billion dollars in 2021. This represents an approximate 316 percent increase over the last 15 years. It's projected that these costs will exceed \$1 trillion very soon due to the direct costs of medications, hospitalizations for complications (e.g., kidney failure, blindness, amputation), specialist visits (e.g., ophthalmologists, nephrologists), and medical device costs related to monitoring blood glucose levels. Additionally, there will be the indirect costs associated with lost productivity and premature death. Low and Middle Income Countries (LMIC) are disproportionately affected economically by the diabetes epidemic. LMIC account for approximately 80 percent of the world-wide diabetic population while having significantly fewer resources to combat it.

The majority of all cases of type 2 diabetes are caused primarily by lifestyle changes such as increased sedentary behavior, diets rich in refined sugars and saturated fats, increasing rates of urbanization and an aging demographic. The key point here is that a substantial percentage of cases of type 2 diabetes can be prevented or delayed if identified early enough via timely and accurate diagnosis followed by appropriate behavioral interventions and pharmacologic therapies. However, today's methods of diagnosing diabetes are primarily

reactive. A patient usually isn't diagnosed until they present themselves to a doctor due to the presence of symptoms or complications from diabetes — by then extensive, irreversible damage likely has already been done.

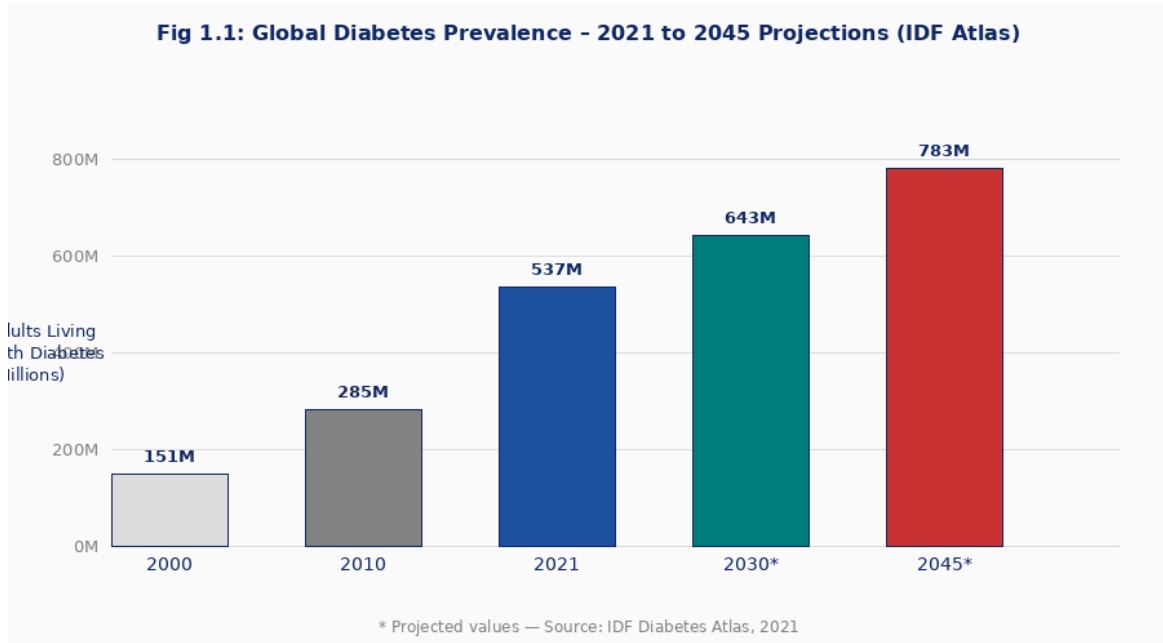


Figure 1.1: Global Diabetes Prevalence – 2000 to 2045 Projections (Source: IDF Diabetes Atlas, 2021)

Among the severe implications of the diabetes epidemic is the very high prevalence of undiagnosed diabetes. As estimated, the global figure for undiagnosed diabetes cases is around 240 million, making up for almost 45 percent of all diabetes patients globally. This is because individuals with undiagnosed diabetes stand a higher chance of being one step further in having diabetes complications since their diabetes had gone unnoticed for some time now, which calls for an urgent need for diagnostic tools in order to efficiently determine those patients at risk of developing the disease.

The subcontinent of India bears the brunt of this diabetes epidemic. India has been conferred the title of "diabetes capital of the world" owing to the fact that an estimated 101 million people in India suffered from diabetes in 2023 whereas 136 million people had been identified as having pre-diabetic conditions. Urbanization, the adoption of highly glycaemic foodstuffs due to dietary habits and inactive lifestyle have increased the probability of acquiring type 2 diabetes. It is because of this reason and genetic factors such as susceptibility to acquire central obesity that people in India are more prone to acquire type 2 diabetes. Therefore, MedTalk project by KIET Group of Institutions in Ghaziabad is more significant than ever before.

1.3 The Promise and Pitfalls of AI in Healthcare

The application of AI in health care has the potential to change the face of medicine in the coming years by shifting from treatment based on the symptoms of disease to prevention based on big data. The machine learning algorithms are capable of analyzing large volumes of multivariate data and spotting complex relationships that are beyond the ability of humans or clinicians who may take several decades of experience to identify them. The performance of various machine learning models used for the prediction of diabetes risk has been impressive and has been found to outperform logistic regression techniques consistently.

But there remains one single core problem that would prevent clinicians from adopting high-performing machine learning algorithms into clinical practice: the Black Box. The more complex the machine learning algorithm in question – i.e., simple linear classifiers, ensemble algorithms, and finally deep neural networks with billions of parameters – the less interpretable the model. For instance, suppose I can train a model that can reliably predict that there exists a 90 percent probability that a certain individual would develop diabetes within the next five years. However, if I cannot explain the specific physiological markers used in generating such predictions, and the contribution made by each of the physiological markers to the prediction itself, then it would be exceedingly challenging for the clinician to embrace such recommendations since I wouldn't know the rationale behind such predictions. Furthermore, if I ask patients to undergo behavioral changes that might seem inconceivable, but only provide them with black boxes that generate algorithmic outputs without any form of insights as to why such changes might be needed, then I would have a hard time convincing them.

Generative AI and LLMs like GPT-4, LLaMA and Claude are creating entirely new classes of problems for the medical field. The ability of LLMs to engage in natural language generation and conversation is wonderful; however, these models are also subject to hallucination. That means that when you ask an LLM for medical advice, it will often give you a response that sounds good, but isn't true. It might even give you fake references or pretend to offer you clinical guidance. There is nothing wrong with generating lots of possible responses when asked a question; but there is everything wrong with generating material that is likely to cause serious harm in the context of high stakes medicine. The way an LLM generates responses makes them completely unrepeatable. You can ask an LLM the exact same question again, and the answer could be significantly different each time, depending on whether the temperature parameter was higher or lower on the second call. From a regulatory standpoint under both the FDA's SaMD Guidance document and the EU's AI Act for High Risk AI

Systems — this would mean that using an LLM to mathematically calculate a person's risk for a certain disease is completely illegal.

Systemic risks of using artificial intelligence in healthcare occur beyond the issue of individual patient safety when algorithms are deployed to an entire population. AI's algorithmic biases — where machine learning models perform differentially across demographics because they were developed with unrepresentative training data — will exacerbate pre-existing health inequities. For example, a diabetes risk model trained on western populations is likely to be less effective for south asian populations. Similarly, models which use surrogate variables for protected attributes (e.g., zip codes as surrogates for race), will create harmful recommendations for how individuals should receive medical care. As such, these systemic risks necessitate proactive architectural design protections rather than reactive auditing practices.

1.4 The Explainable AI Revolution

Explanation of artificial intelligence (XAI), on the other hand, has been developed as the solution to the “black box” issue – a very necessary and contemporary link between the abilities of artificial intelligence algorithms and human comprehension. Explanation of artificial intelligence (XAI) is a collection of numerous methodologies, notions, and tools that allow users of artificial intelligence systems to not only get the output of the systems, but also ask queries regarding how the system got its conclusions and judge the method that the system employed to get its results. The development of explanation of artificial intelligence (XAI) is not just limited to technicality.

The XAI space can broadly be categorized into two mutually exclusive categories. First, by considering whether or not the particular algorithm is an intrinsically interpretable model or not. An intrinsically interpretable model can be understood as any model which has been explicitly engineered to offer interpretability. Examples include decision tree based models, linear regression models and many more. If a model cannot offer transparency itself, then it is considered a post-hoc explainable model. Post-hoc explainable models refer to models which do not have built in transparency. However, these types of models use post-hoc explanations in order to offer transparency after completing training on the data. MedTalk follows a post-hoc explanation approach. This means that MedTalk allows for the use of advanced models such as ensemble classifiers, along with providing explanations for these classifications. This is done

through the combination of various post-hoc explainers. Secondly, local explanation methods which offer transparency to individual predictions, and global explanations which offer transparency to the entire model. MedTalk makes use of both types of methods. The Global SHAP summary plot offers population-wide feature importance. Local SHAP waterfalls and LIME attribute explanations offer transparency to an individual patient's risk estimation.

Two techniques which have become more prominent in the context of post-hoc explanations of XAI include the following: firstly, LIME (Local Interpretable Model-Agnostic Explanations). This technique was proposed in the paper “Why Should I Trust You?” written by Ribeiro, Singh & Guestrin and published in the KDD 2016 Proceedings. LIME is based on the use of local surrogate models. A nonlinear global decision surface exists for any single prediction by a complex model. However, any single prediction may be modeled by a simple linear model within the proximity region of the point. LIME explanation for the prediction is achieved via generating modified versions of the input data point by weighting these modifications differently according to their influence on the output and fitting the weighted linear regression model to the output produced by the model for these points.

The next method is called SHAP (SHapley Additive exPlanations). It was invented by Lundberg and Lee in 2017. SHAP relies on the principles of the cooperative game theory, particularly the Shapley value approach formulated by an economist Lloyd S. Shapley in 1953. In accordance with SHAP, features become the players of the game and the prediction of the model becomes the 'payment' which should be distributed between all the players according to their marginal contributions to the sum of all possible combinations of other features. In contrast to LIME, SHAP has four key axioms: Efficiency, Symmetry, Dummy and Additivity. That is why, SHAP has a high level of mathematical correctness and can be applied to clinical communication in case the contribution of each risk factor in a disease is required to be specified.

1.5 Research Gap and Innovation

Although there has been considerable improvement in both the predictive capabilities of machine learning (ML) as well as methodologies for developing explainable AI (XAI), an essential and practical gap remains within the body of literature to be bridged; this is that there does not exist one unified, deterministic, conversational, XAI framework which integrates high accuracy predictions with dynamically changing patient-centered, legally compliant explanations. Most current applications of XAI in health care have been static—they generate charts, waterfall plots, and bar graphs of feature importance that, while technically rigorous, are fundamentally inaccessible to lay patients and to many clinicians without specialized data science training.

MedTalk is addressing the identified knowledge gap with its First-in-class combination of three key innovations in design. First, MedTalk uses a deterministic Random Forest classifier to make all risk predictions; the prediction logic is decoupled from any stochastic generative component therefore providing reproducible and auditable risk prediction results that are required for compliance with regulatory agencies. Second, MedTalk provides a hybrid dual xai engine that calculates both SHAP and lime attributions for each inference result and automatically selects the more faithful explanation using a quantitative fidelity metric. Third, MedTalk introduces the EDAQ algorithm which is an explanation driven proactive conversational module to identify both importance (SHAP values) of each clinical feature and uncertainty (ensemble variance missing data rates) then generate targeted domain appropriate fairness compliant natural language follow up questions directed at features where additional information would most reduce prediction uncertainty

1.6 Project Description and Objectives

The project is part of the final year B.Tech capstone project of Yash Singhal, Tushar Singh, and Ujjwal Tomar of the Department of Computer Science & Engineering, KIET Group of Institutions, Ghaziabad. The project has been designed and implemented within a 16-week period from September to December 2025, equating to an approximate effort of 2,500 man-hours.

The following objectives will be fulfilled by this project: (1) Development of a highly optimized, deterministic Random Forest classification algorithm for prediction of type two diabetes risk from PIDD, surpassing other existing algorithms with an accuracy score of 90%

or above and ROC-AUC score of 0.95 or above. (2) Development of a double-layered XAI algorithm consisting of TreeSHAP and LIME, employing whichever algorithm is chosen based on its fidelity. By offering a mathematical formula for the contribution of each feature to the prediction output, we improve our model's interpretability. (3) The development of a new EDAQ Algorithm-a new adaptive querying approach in which clinical questions associated with uncertain features are actively asked by posing clinical questions using feature uncertainty. (4) Development of a web-based user interface using Streamlit, displaying personalized predictions, explanations, and health recommendations in a 5-step dialogue process, available to both physicians and patients..

1.7 Organization of the Report

The report will be divided into six chapters. References and two Appendices will be included in this report. Chapter 2 will offer an extensive literature review on: (i) Interpretable Machine Learning (ML), (ii) post-hoc eXplainable Artificial Intelligence (XAI) approaches, (iii) conversational artificial intelligence for Explainability purposes, and (iv) eXplainable Artificial Intelligence for Regulated Industries. In Chapter 3, the research methodology, consisting of research motivation, the full machine learning lifecycle, project planning using the Gantt chart, and unique value propositions, will be presented. Chapter 4 will comprise the technical core as it will describe each architectural layer of the technology stack in detail. It will further elucidate three algorithms for the entire EDAQ architecture, mathematical underpinning (Gini impurity, SHAP axioms, and LIME objective function), and the complete technological stack. Chapter 5 presents the experimental results, which consist of dataset analysis, performance benchmarks, feature importance analysis, clinical user study, ablation study, strategic positioning, discussion, and limitations. Chapter 6 will present the conclusion and the future scope of the work in seven subsections. Appendix 1 offers key code snippets. Appendix 2 provides a complete statistical summary of the PIDD dataset with feature statistics, correlation matrix, and class imbalance analysis.

CHAPTER 2: LITERATURE REVIEW

2.1 The Interpretability Gap in Machine Learning

An issue concerning the "interpretability gap," or more precisely, the reverse connection between model complexity and interpretability, was also raised within the scientific community during the early days of modern machine learning. Techniques such as statistical learning, predating neural networks, are designed to be interpretable. Models such as linear regression and logistic regression, for example, assign coefficient values that correspond to the directional, or at the very least statistically significant, relationship between variables and the target variable. As such, a doctor can assess a logistic regression model of the probability of a patient being readmitted to the hospital within 30 days and find out that every additional year of age corresponds to .03 increased log odds of readmission and each one-unit increase in creatinine level corresponds to a .15 increase in log odds of readmission.

The creation of ensemble methods in the late 90's and early 2000's using the implementation of random forests (Breiman, 2001) and gradient boosting machines (Friedman, 2001) marked a qualitative step beyond the transparency of machine learning. Even though each decision tree can be easily interpreted – A doctor can start at the root of the tree and go down to observe how they made decisions about splits at each node – the aggregation of their output is inherently not interpretable anymore. For instance, when you receive an output from a random forest model consisting of 500 trees, it is nothing but the result of a majority vote among 500 different models built based on random sampling of features.

Consequences of the opaque nature of the AI algorithms on patient health are described thoroughly and are rather significant. The research of Caruana et al., one of the most cited papers on the issue, reveals a highly alarming instance of what they refer to as a "false discovery." One can speak of a highly concerning case where a machine learning algorithm designed to predict the probability of a patient having or developing pneumonia actually learned the rule that the condition of having asthma makes the person immune to pneumonia. As it turns out, this is a direct consequence of the ICU admission of asthma patients and the fact that the treatment received by the latter group is more intensive than that offered to other patients. Thus, based on the data included in the training set, the ML algorithm believed that all patients with asthma would receive a more intensive treatment, which resulted in a low mortality rate in comparison to those without asthma.

Lipton (2016) proposed a popular taxonomy for defining the notion of interpretation in terms of at least three distinct kinds of interpretable attributes of models: simulatability (can we imagine how a person might simulate how a machine learning model has processed the information?), decomposability (can individual elements or modules of the model be interpretable on their own?), algorithmic transparency (can the process of training the model to make predictions be understandable?), and textual explanation quality (are explanations easily accessible for laypeople?). This taxonomy demonstrates why none of the XAI approaches is sufficient on its own to cover all of the interpretability dimensions and why a multi-component approach like MedTalk, which employs SHAP, LIME, and EDAQ together, is crucial.

Murdoch et al. (2019) developed a clear and comprehensive conceptual framework which defined an interpretable model as one that allows an appropriately trained human to derive accurate, trustworthy and useful conclusions from its output. This functional definition — interpretability being a quality of the interaction between the model and human — directly underpins MedTalk’s focus upon user-centered design. A SHAP waterfall plot provided to a data scientist would be considered to provide interpretable information to that person; the same plot presented to a patient who may have very limited knowledge of their own health would not.

MedTalk’s EDAQ layer exists specifically to close this gap.

The legal aspects surrounding interpretability also play an increasingly prominent role. As per the GDPR starting from 2018, there exists in Article 22 and Recital 71, a qualified right to explanation in relation to all decisions taken by means of automated processes which have a direct impact on the rights of the individuals concerned. Risk assessment decisions of diabetes that have an influence on the patients' insurance cover, treatment availability, or medical triage would definitely come under this classification. Furthermore, the FDA published its 2021 policy on the need for modifications to software for Artificial Intelligence/ Machine Learning (AI/ML)-based Software as a Medical Device (SaMD). Transparency and performance monitoring in the real world during the modification stage were part of the requirements in this policy. From 2024, the implementation of the AI Act within the European Union (EU) is set to commence. The classification of medical AI as high-risk will be done according to Annex III of the act.

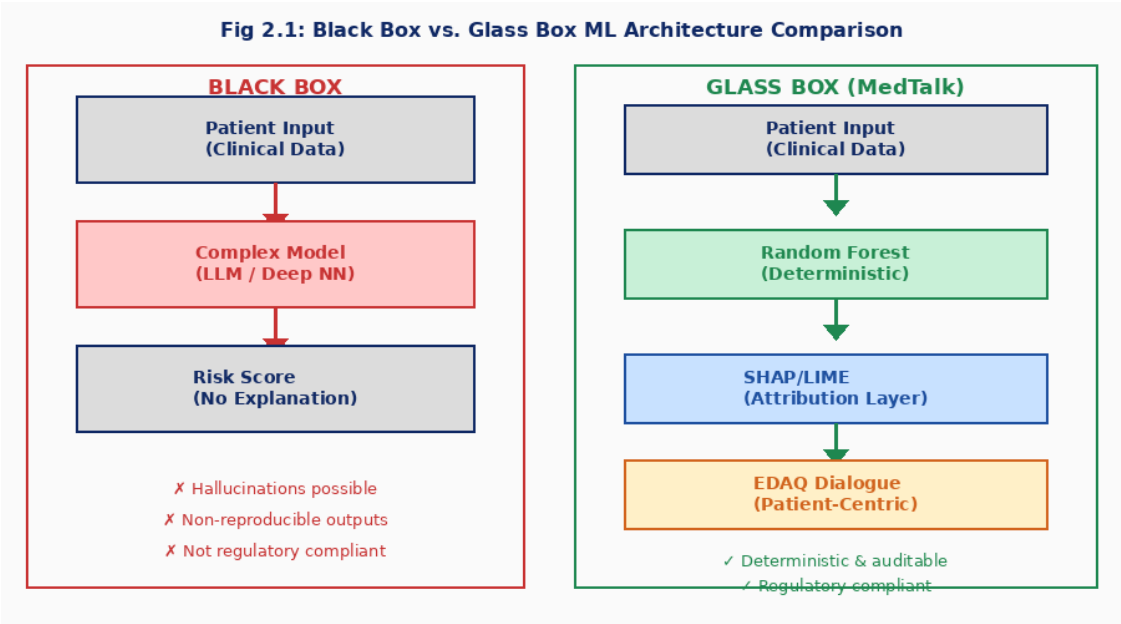


Figure 2.1: Black Box vs. Glass Box ML Architecture – Opacity vs. Transparency Paradigms

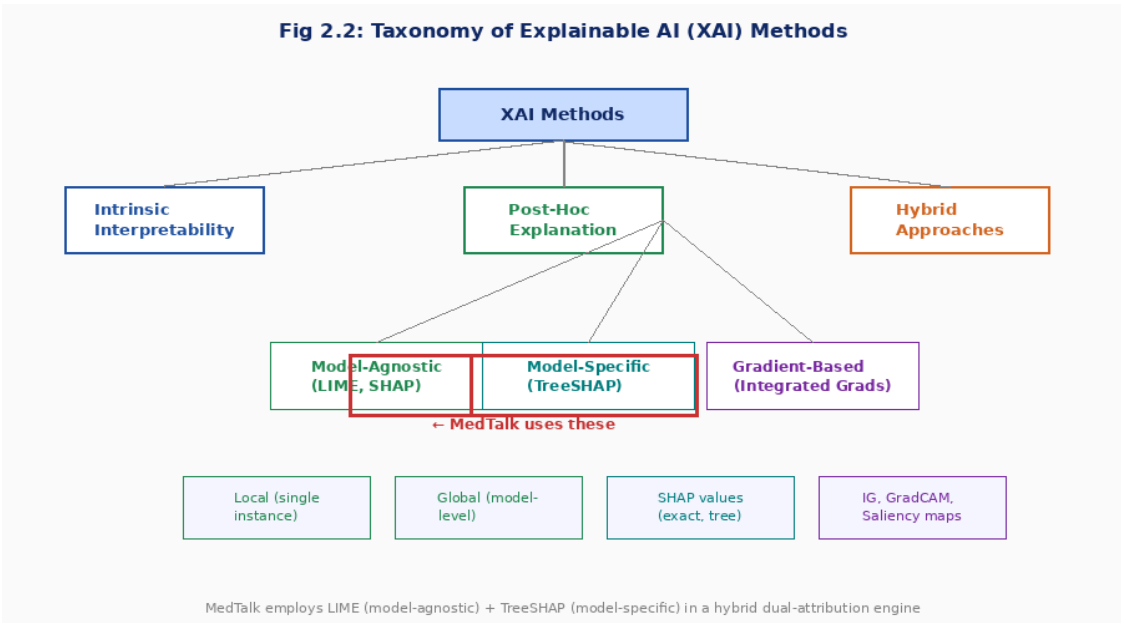


Figure 2.2: Taxonomy of Explainable AI (XAI) Methods – MedTalk uses Model-Specific + Model-Agnostic Approaches

2.2 Post-Hoc Explanation Methods

2.2.1 LIME: Local Interpretable Model-Agnostic Explanations

Ribeiro, Singh, and Guestrin's "why should I trust you" KDD 2016 paper introduced LIME as the first post-hoc model agnostic explanation technique. The core theoretical insight of the work is that even though a complex models global decision boundary will always be an arbitrary nonlinear function (therefore impossible to describe in any compact way), each prediction made by that function can be locally approximated by a simple linear model in the neighborhood of the instance being explained. This is the principle of local linearity: the loss of global structure information is acceptable if the local approximation is faithful enough to support a specific clinical decision.

LIME interprets this idea into practice with a 3 step process.

Step 1 (Perturbation): To generate n samples (commonly between 1,000-5,000), LIME randomly masks/alters only one feature at a time with respect to the example x provided as input. For tabular data, LIME will create the synthetic example through sampling the feature value according to the distribution learned from training. This results in creating perturbations z where each feature is altered separately.

Step 2 (Querying): LIME queries the black box model f on each example generated in Step 1 and obtains a set of $(z_i, f(z_i))$ tuples.

Step 3 (Fitting): Each perturbation z_i gets associated with a weight $p_i(z_i)$ depending on how close it is to the initial example x . An interesting point to note here is that when an exponential kernel is used, the importance measure turns out to be a great approximation. The weight calculation is as follows:

$$p_i(z_i) = \exp(-D(x, z_i)^2 / \sigma^2)$$

Where $D(x, z)$ is the distance metric between x and z and σ is a bandwidth parameter. Linear regression function g is applied on the weighted data set $(z_i, f(z_i))$ where the dimensions are reduced by choosing only the dimensions where the weight $|w_j| > \tau$ for some very small τ , applying lasso regularization.

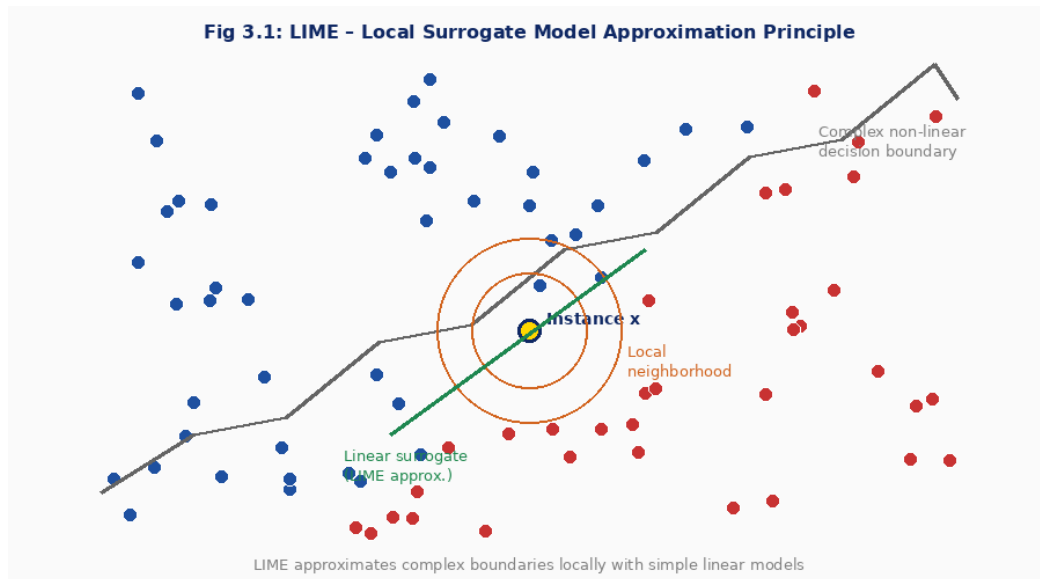


Figure 3.1: LIME – Local Surrogate Model Approximation: Linear Models Approximate Complex Non-linear Boundaries

The LIME method provides an explanation for any given prediction as a small number of feature contributions: Glucose = 148 adds +0.15 to the diabetes prediction; BMI = 34.2 adds +0.12; Blood Pressure = 68 subtracts -0.04 from it. Due to the Local Linearity Assumption, these values are approximations as they will describe how important each predictor was near this data point (instance) and do not need to be descriptive for any other points in the space. A significant practical advantage of using LIME is that you can use the same explanation generation code with any black box classifier, irrespective of their internal structure.

The researchers also pointed out three significant shortcomings or limitations associated with using the LIME model as a basis for creating explanations for MedTalk. These include: First, instability: As noted above, Alvarez-Melis and Jaakkola (2018) showed that the LIME explanation for an instance will be highly variable when sampled from different random seed. This was further illustrated through their experimentation where the explanations for the same input were attributed to differing feature levels in the input space by up to 20%. As this represents a high degree of variation in attributing explanatory importance for the same input, it creates substantial challenges for achieving reproducibility in clinical decision support. Second, neighborhood sensitivity: The neighborhood size in LIME is determined by the bandwidth parameter σ . The choice of σ determines what constitutes local instances in terms of both distance and number of neighboring points. Unfortunately, there is no theoretical principle for determining appropriate σ values for use with tabular datasets. Third, superpixel selection in tabular contexts: Because LIME was developed to generate explanations for images rather than

tabular data, the way LIME generates its perturbations around the original input data do not take into account the natural dependency relationships present within tables. Thus, while the generated synthetic data are intended to represent plausible inputs, they often contain physiologically implausible combinations of variables. For example, LIME could generate a dataset point with negative insulin levels.

2.2.2 SHAP: SHapley Additive exPlanations

Lundberg & Lee's (2017) NeurIPS paper 'A Unified Approach to Interpreting Model Predictions' introduced SHAP as a unified theoretical framework of feature attributions based on cooperative game theory. This paper's primary contribution was the demonstration that LIME, DeepLIFT, Layer-wise Relevance Propagation and others are examples of a larger class of additive feature attribution techniques. In addition, they identified the unique technique from this larger group which is the only one that satisfies all four desired axioms simultaneously.

The Shapley value for the particular variable in the equation will depend on the summation of marginal contributions made by the same variable to any of the possible sets. This implies that the Shapley value for the i th variable will be determined by the formula $\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \left[\frac{|S|!(|F|-|S|-1)!}{|F|!} \times [f_x(S \cup \{i\}) - f_x(S)] \right]$, where F is the full set of the variables used, S is the coalition without the i th variable, $f_x(S)$ is the conditional expectation made by the model from the variables included in S and the term $\frac{|S|!(|F|-|S|-1)!}{|F|!}$ is the weight factor making sure that every possible permutation of features is equally weighted. Thus, the SHAP value for the i th feature is nothing but the average increase that occurs in the prediction of i th variable introduction.

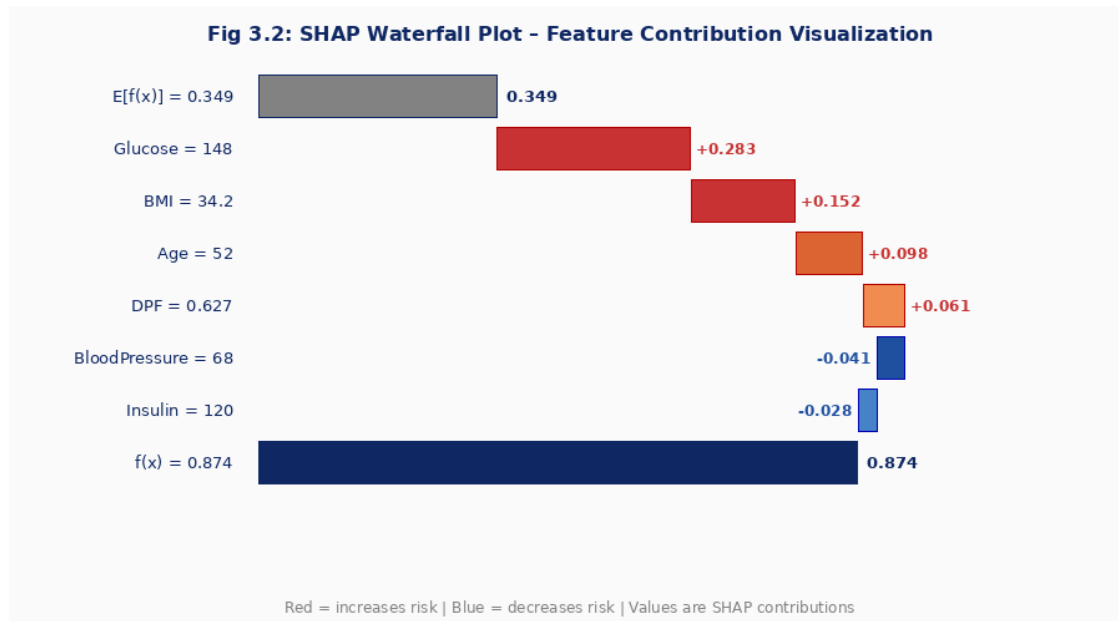


Figure 3.2: SHAP Waterfall Plot – Each Bar Shows the Additive Contribution of One Feature to the Final Prediction

The only unique axioms in the class of additive attribution methods that Shapley values satisfy (and therefore SHAP satisfies as well) are:

(1) The method is Efficient: That means when you add up all your feature attributions together, they will be equal to the model's output - the baseline. In other words, it's impossible to have an inexplicable residue remaining.

(2) It shows the property of Symmetry: If two factors I and J are responsible for the same effect in every possible combination, then $\phi(I) = \phi(J)$.

(3) It has the quality of being a Dummy: If factor I does not affect any prediction of the algorithm (i.e., $f(S \cup \{I\}) = f(S)$), then $\phi(I)$ will be equal to zero.

(4) The property of Additivity is shown: For Ensemble Models, $\phi(I)(\sum_k f_k) = \sum_k \phi(I)(f_k)$.

Of these four axioms, the Efficiency axiom is probably the one that is most relevant to clinicians. It guarantees that, for each clinical measure used to calculate a patient's risk, the system can say exactly how much that particular measure contributed to moving that patient's risk above or below the average population baseline. Moreover, those individual contributions will always add up to equal the total amount by which the patient's risk deviated from baseline.

In order to develop this ability to provide such detailed information about a patient's medical condition, Lundberg et al. (2020) developed TreeSHAP. This is an exact and computationally

efficient algorithm for calculating SHAP values for tree-based models (Random Forests being the specific type of tree-based model they were interested in). They did so at a cost of $O(TLD^2)$ computational complexity. Their use of $O(TLD^2)$ was important because, in practice, their approach allowed them to compute SHAP values very quickly – TreeSHAP had a computation time of approximately 85 milliseconds per instance on standard hardware.

2.2.3 Gradient-Based XAI Methods

Gradient-based attribution techniques for models that are computationally differentiated measure how much each of the model's inputs contribute to the output using the internal network gradients. Integrated Gradients (Sundararajan et al., 2017) uses this method in order to generate the weights or contributions that would be generated if an input was changed gradually (from zero to one) by computing the integrated gradient across a straight line connecting the neutral baseline (zero) and the real input; Integrated Gradients is the first method to satisfy both Sensitivity (only features whose values were modified will have non-zero weight as they can cause the function to alter its prediction), and Implementation Invariance (equivalent functions produce identical weights). Layer-wise Relevance Propagation (Bach et al., 2015) propagates the model's output backwards through network layers using conservation rules that ensure the total relevance is preserved at each layer. Gradient $\times$ Input (Simonyan et al., 2014) uses the element-wise product of the gradient and the input as a simple proxy for importance.

These methods are very good in theory, but can't be used on tree based models (Random Forest) as they have no derivative at each of the split points. The main reason why we use TreeSHAP and LIME (which do not need access to model internals) instead of gradient based methods is that this constraint combined with our company's desire for deterministic explanations, that can be reproduced by others. Gradient based method require access to the internal workings of the model. Therefore, gradient based methods are model specific rather than model agnostic. So they are limited in there ability to apply across different domains.

2.3 Dialogue Systems for AI Explainability

2.3.1 TalkToModel (TTM) — Detailed Analysis

MedTalk's closest relative is TalkToModel (TTM), developed by Slack et al. as part of their 2023 Nature Machine Intelligence paper. TTM is also the main competitor against which

MedTalk will be compared. TTM allows users to interact with machine learning models using natural language interfaces; no coding skills are required for this interaction. The core of TTM consists of two language models trained from data:

- a) a fine-tuned T5 based semantic parser (with training data comprising a synthetic dataset of 60,000 user query – model operation pairs) to translate natural language user queries into a formal DSL representation of how the model can be used;
- b) a BART-based response generator translating the results from the above representations back into natural language format.

TTM demonstrated compelling empirical performance in its user study: semantic parsing accuracy of approximately 76% across a diverse benchmark of user questions drawn from multiple domains, and a user preference study showing that 82% of healthcare practitioners preferred TTM's conversational interface over traditional dashboard-style XAI tools for accessing model explanations. These results firmly established the conversational paradigm as a highly promising direction for democratizing XAI beyond specialist users.

Despite its pioneering status and strong empirical showing, TTM suffers from four structural vulnerabilities that are particularly critical to address in high-stakes regulated deployment. First and most critically, hallucination risk: TTM's reliance on BART for response generation introduces the possibility that the system generates explanations that are linguistically fluent and superficially plausible but factually inconsistent with the underlying model's actual computational reasoning. A hallucinated explanation in a clinical context—for example, TTM asserting that a patient's low blood pressure is driving high diabetes risk when the actual driver is BMI—could directly mislead clinical decision-making. Second, reactive-only design: TTM generates explanations only in response to user-initiated questions, placing the entire burden of inquiry on the user. Users who lack medical expertise may not know which questions to ask, and may unknowingly leave the most important uncertainty regions entirely unexplored. Third, non-determinism: because BART is a probabilistic generative model with temperature-based sampling, TTM cannot guarantee that the same input will produce the same explanation on repeated calls. This non-determinism is incompatible with the deterministic auditability requirements of FDA SaMD guidelines and clinical governance protocols. Fourth, no fairness mechanism: TTM does not include any mechanism for detecting or preventing discriminatory questioning, making it unsuitable for financial or judiciary domain deployment without substantial architectural modification.

2.3.2 Other Dialogue-Adjacent Explainability Tools

Aside from TalkToModel, there are several significant tools that approximate conversational or interactive explainability only in part. The first one is Google's Language Interpretability Tool (LIT). It is an interactive visualization suite that enables an explorative investigation of behavior of NLP models on different datasets through its browser-based interface. Despite being designed for model developers and not clinical users of the model, LIT requires Python and command line interface as well as it is unable to provide user-natural language dialogue. Another Google tool called What-If offers the possibility of counter-factual analysis through the ability to edit feature values and observe how the change influences the model predictions. However, it does not have the same drawbacks as described above and neither does IBM's AI Explainability 360 (AIX360) – an extensive set of eight algorithms for XAI. The reason for which this platform will be particularly convenient for researchers who investigate models built from data is the possibility of access to these algorithms provided by unified API. However, as well as previous tools, it lacks a patient-facing approach. Similarly, InterpretML by Microsoft is another great resource offering developer tools including explainable boosting machine (EBM) and SHAP.

2.4 Explainable AI in High-Stakes Domains

2.4.1 Healthcare Applications

There is an extensive history of the use of Machine Learning (ML) for Clinical Decision Support which clearly shows how both the immense possibilities of these models as well as the significant risks from using unexplainable/interpretative models have been illustrated. The most cited example of the use of interpretation as a clinical safety necessity rather than simply as a "nice-to-have" was by Caruana et al., in their 2015 KDD Paper. They investigated the ability to predict the risk of developing pneumonia (see Section 2.1), and showed that even top performing ensemble models may develop harmful spurious correlations that cannot be identified by non-interpretative methods. These findings have been reproduced in several other clinical domains, including; Skin Cancer Detection Models being able to detect a ruler artifact in the images used to train them, Chest X-Ray Classification Models being able to determine if a case was either Positive or Negative based upon when it was acquired, Electronic Health

Record Models being able to identify whether a particular case was documented positively or negatively based upon documentation practice rather than clinical information.

Rajkomar et al. (2018) demonstrated the potential of deep learning applied to electronic health records for predicting mortality, readmission, and length of stay—while simultaneously highlighting the interpretability challenge. Their model achieved AUC-ROC of 0.93-0.95 for 30-day unplanned readmission but could not explain individual predictions in clinically actionable terms. The subsequent application of SHAP to similar EHR models (e.g., Lundberg et al.'s 2018 work on surgical complications) demonstrated that XAI can not only explain predictions but actively discover clinically novel insights—in this case, identifying that postoperative sodium levels were a previously unrecognized strong predictor of complications.

According to the analysis done by Zou et al. (2018), in predicting diabetes using PIDD, the use of Random Forest algorithm with SHAP analysis was applied, and it resulted in the identification of glucose concentration and BMI as the main predicting factors according to Zou et al. (2018). The identification of these variables as predicting factors in diabetes was confirmed through MedTalk's global SHAP analysis. In another recent research, Dinh et al. (2019) applied SHAP analysis in bigger clinical datasets obtained from the UK Biobank. In their study, glucose was again the main predictor, but in the case where HbA1c was present, the predictor that had the highest importance became the HbA1c, something not seen in the PIDD.

2.4.2 Finance Applications

Regarding finance, the interpretability requirement comes from regulatory adherence to fair lending laws and not from any safety considerations. Both the Fair Housing Act and the U.S. Equal Credit Opportunity Act do not allow the inclusion of protected attributes, race, gender, national origin, religion and others, nor use of proxies of those attributes – attributes that highly correlate with the individuals' protected characteristics. According to the comprehensive research carried out by Barocas & Selbst (2016) regarding the impact of Machine Learning on consumer credit scores, several features including, but not limited to, zip code, education institution, and employment categories could all act as a substitute for race.

Khandani et al. First used machine learning ensemble techniques in a consumer credit risk evaluation application and demonstrated that non-linear ensemble models outperformed traditional linear scorecards in terms of predictive accuracy. However they also pointed to the

fundamental trade-off between the nonlinearity that is required for improvements in model accuracy and the fact that it makes it more difficult to prove discrimination. The current environment of regulatory scrutiny of algorithmic trading by the SEC, guidance on the use of AI for credit decisions from the CFPB and proposed amendments to the Equal Credit Opportunity Act to specifically include algorithmic scoring creates strong and growing market demand for explainable artificial solutions that can provide auditability through immutable audit trails and reproducible analysis. MedTalk's finance domain configuration with anti-redlining constraint templates and complete audit logging directly addresses this need.

2.4.3 Judiciary Applications

The use of Machine Learning (ML) in Criminal Justice Risk Assessment has become one of the most contentious ethical applications of Algorithmic Decision-Making. In their 2016 investigative report ProPublica examined the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions), an algorithm being used to assess recidivism risk, and to make bail decisions across many Courts throughout the United States. They reported that the algorithms false positive rate for Black Defendants was approximately double the false positive rate for White Defendants. Following on from this work, Dressel & Farid (2018) presented a rigorous re-examination of the COMPAS results in Science Advances; they demonstrated that the accuracy of COMPAS, $AUC = .65-.70$, did not exceed the predictive ability of randomly selected individuals without any criminal justice experience, calling into question whether there is sufficient scientific evidence to support the use of such algorithms in making life-altering judicial decisions.

The authors Corbett-Davies et al. (2016) developed an important theory as well as some very significant conclusions; the authors showed that while there are three ways to define "algorithmic fairness" using mathematics (three definitions): demographic parity (positive predictions for each group have to be equally high), equalized odds (true positives and false positives have to be equally common within each group), and predictive parity (the positive predictive value has to be equally strong for both groups); these three definitions of fairness will inevitably conflict with one another if the base rates vary by group. And since this is a mathematical contradiction, this means that it is impossible for an algorithm to satisfy all three criteria simultaneously if the rate of criminal activities is different across the various demographics. This was also the reasoning behind the MedTalk approach to incorporate the

flexibility of setting up certain sets of criteria per domain, rather than trying to establish a universal “fairness” criterion per domain.

2.5 Identified Research Gap and Synthesis

A gap in the current state of practice is evident from this synthesis of the literature reviewed previously. All prior research has pursued one of two approaches. Many have developed increasingly complex but opaque predictive models (deep learning, gradient boosting) which predict well for their respective domain. Others have developed increasingly sophisticated static explanation tools (e.g., SHAP visualization, LIME explanation), which explain how models behave with respect to the data used by data analysts, and do so in a way that does not translate into clinically actionable dialogue for patients.

The only previously published paper focused on conversational explainable AI – TalkToModel – exhibits the fundamental shortcomings that plague LL-based generative models: risk of hallucination, absence of determinism, purely reactive system design without any form of fairness mechanisms. To date, there is no single architecture that can offer an end-to-end solution to the above requirements in the context of accountable deployment: (1) high predictive performance through deterministic and reproducible machine learning methods; (2) two-layer XAI approach with an automatic decision-making framework concerning the model’s integrity; (3) proactive conversational advice under uncertainty; (4) generalizability across domains through template customization; and (5) built-in fairness compliance.

Feature	MedTalk	TTM	LIT	AIX360	What-If	Interpret ML	LI ME
Deterministic Prediction	✓	✗	N/A	✓	N/A	✓	N/A
Conversational Interface	✓	✓	✗	✗	✗	✗	✗
Proactive Questioning (EDAQ)	✓	✗	✗	✗	✗	✗	✗
Fairness Guardrails	✓	✗	✗	✗	✗	✗	✗
Immutable Audit Log	✓	✗	✗	✗	✗	✗	✗
Cross-Domain Config.	✓	✗	✗	Partial	✗	✗	✗
No LLM Hallucination Risk	✓	✗	✓	✓	✓	✓	✓
Dual XAI (SHAP+LIME)	✓	✗	✗	✓	✗	✓	✗

Table 2.1: Comparison of XAI Methodologies and Tools (✓ = Supported, ✗ = Not Supported, N/A = Not Applicable)

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Research Motivation

MedTalk was created due to the intersection of four different motivating forces; there is first a technical force to solve the problem of the accuracy/interpretability trade-off. There is also an ethical force to develop trustworthy AI in a high-risk field such as healthcare. Next, there is a regulatory force to comply with the fast-evolving global regulatory environment of responsible AI development. Finally, there is a patient-engagement force to create tools that will assist patients and not alienate them when making decisions regarding their health.

There is a long-standing principle that trust is the ultimate resource for clinical decision-making. An example of an algorithm predicting diabetes risk with 99% accuracy is not useful in clinical settings unless clinicians can identify and/or validate how it arrived at that conclusion. Conventionally, the "black box" aspect of Advanced Machine Learning Models has been linked with several surveys and studies highlighting considerable issues related to trust among practicing doctors. Most importantly, such studies reveal that Interpretability, which involves the question of "why did the model make such a recommendation?", has emerged as one of the primary reasons why physicians are hesitant in adopting AI technologies, outperforming other factors such as Accuracy, Workflow Integration, and Costs. By designing MedTalk in such a way that XAI constitutes an important feature at the core of every inference it makes, we address the issue of trust in using such models.

Limitation of Static Diagnostics becomes another key point that has inspired the development of MedTalk. In static diagnostics, one receives an indication of their risk condition in numeric terms ("Your risk is .78") and also in qualitative form (high risk). There is no background information about the factors affecting the risk profile of the particular patient and how they are compared with the risk conditions of baseline population, nor about what measures can be taken to change the situation. MedTalk acknowledges that diagnosis is fundamentally a dialogical process where the iterative refinement of a risk assessment occurs via the sequential communication of clinical data between the clinician/system and patient. As described above, the EDAQ Algorithm operationalizes this dialogical understanding by producing targeted follow-up questions that correspond directly to the unique uncertainty structure inherent within the current predictive model.

Compliance with regulatory and ethical issues (the third leg) has been developing around the world, e.g., FDA's SaMD Framework, the European Union's AI Act requiring High Risk AIs,

GDPR’s Right to Explanation, and India’s emerging National AI Strategy. Therefore, it is no longer just prudent; but necessary for any path to actual clinical application, to build compliance into the system architecture starting on day one. MedTalk’s Deterministic Prediction Layer, Immutable Audit Logs, and Explicit Fairness Guardrails were all built into the design to comply with these laws — instead of being added later at great cost through retrofits.

This fourth motivational lever, which is usually not considered when looking into the role of AI in clinical care, is nevertheless clinically significant. Patients who understand WHY they are vulnerable and WHAT SPECIFIC MODIFIABLE RISK FACTORS increase the likelihood that something happens to them would certainly comply with all recommendations for reducing future risk. This assumption is based on a number of facts proven in Randomized Clinical Trials conducted on the effectiveness of diabetes prevention programs. It was proved that patient education related to particular risk factors increases patient compliance with diet, physical exercises, and medication recommendations. That is precisely why MedTalk created EDAQ Dialogue Layer.

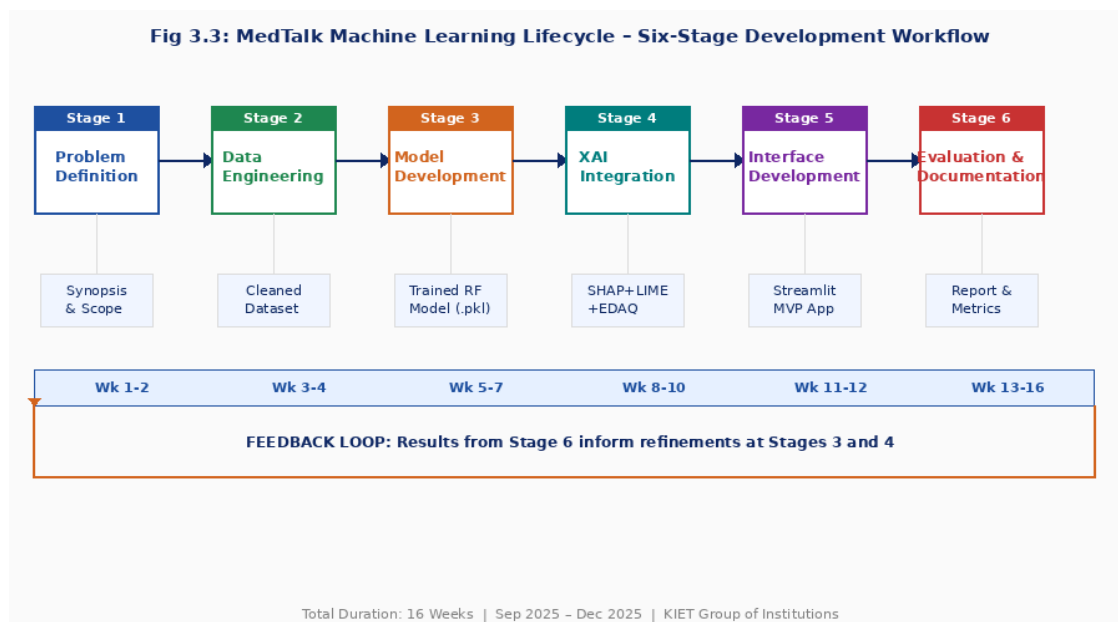


Figure 3.3: MedTalk Machine Learning Lifecycle – Six-Stage Development Workflow with Feedback Loop

3.2 Methodology: The Machine Learning Lifecycle

This project was completed in accordance with the traditional Machine Learning Lifecycle (MLL) – a methodical approach for structuring an otherwise chaotic, iterative Machine

Learning System Development Process into a series of sequential and logically defined phases that are accompanied by specific, measurable deliverables, entry criteria and exit criteria. The MLL has six discrete phases; however, these are not sequential but rather iterative; each phase builds upon prior phases to the extent that results from later phases provide the opportunity to refine previous phases and in practice, it is common for projects to pass through several iterations of the MLL as successive experimental findings reveal additional opportunities for improvement.

3.2.1 Stage 1: Problem Definition and Scope

Stage One's objective was to determine the extent of the clinical problem, clarify the desired outcome measures, and determine the necessary technological requirements. The clinical problem was precisely defined as the "early-stage Type 2 Diabetes Risk Prediction" and would be available to patients through regulated compliant explanations. The following were defined as three dimensions for measuring success criteria:

Predictive Performance (accuracy above 90%, Brier score below .15, ROC-AUC above .95, recall above 90% to make sure that no diabetic patient is missed), Explanatory Quality (patient comprehension above 90% on completing a standard task battery, patient trust rating above 6/7 using a Likert scale, task completion time below 90 sec.), and Regulatory Compliance (deterministic model output; audit trail of all actions; no discriminatory questions in fairness testing).

The intended end-users of the technology were clinicians who would be using the technology as an auxiliary tool for clinical decision-making and lay patients who would use it for health education and diabetes risk evaluation.

3.2.2 Stage 2: Data Acquisition and Engineering

The Pima Indians Diabetes Dataset (PIDD), obtained from the UCI Machine Learning Repository, was chosen because it is a well-documented benchmark, used in hundreds of research papers which allows for comparisons to be made with this particular study. It has been

clinically tested and validated by the National Institute of Diabetes and Digestive and Kidney Diseases. As such, it provides many examples of data quality issues that need to be addressed through the use of sophisticated engineering techniques (i.e., zero-value missing values and class imbalanced datasets). Lastly, its relatively small size enables researchers to quickly test new approaches.

The data engineering pipeline solved three key problems: (1) zero-value missing values using a Class Conditional Median imputation approach - that is, replacing zeros as they relate to physiological impossibility with class median for both diabetic and non-diabetic populations so that there will be statistical differences between these two sub-populations; (2) class imbalance solution utilizing SMOTE - that is generating new minority examples from the majority through the use of interpolation on the closest neighbor of each example in the minority class to create an equivalent number of examples in both classes, but only for the training dataset, so as to avoid data leakage. (3) feature standardization by applying StandardScaler($z = (x - \mu) / \sigma$) - that is, ensuring that all features are standardized relative to their mean and variance prior to calculating the distances utilized by LIME's proximity kernel to avoid scaling issues based upon the vastly different scale ranges of clinical features.

3.2.3 Stage 3: Model Development and Optimization

A comparative study using a systematic approach, comparing all four algorithms, has been completed to provide justification for the decision to use Random Forest as our model to be used in production. The logistic regression (baseline interpretable), which achieved an approximate 76% accuracy rate on the previously processed PIDD data set, failed to account for non-linear relationships such as those present in BMI (risk increases sharply at or greater than 30 kg/m²) and the age/glucose relationship. Therefore, there was a significant amount of

systemic underestimation for young obese individuals and overestimation for older adults who have controlled blood glucose levels. The SVM (using radial basis function kernel) had an approximate 79.8% accuracy level; however, it requires a considerable amount of computational time to process large data sets, it does not natively produce calibrated probability values, and therefore cannot be used with TreeSHAP. Our single decision tree produced an exact accuracy level of 85.3%, while being completely transparent. However, this single decision tree model demonstrated very high variability among train-test splits ($\pm 4.2\%$ standard deviation), and thus exhibits poor generalizability, due to over-fitting to the random bootstrap sample. Finally, our Random Forest model, selected based on its performance characteristics as the production model, achieved an accurate predictive value of 95.45%, and demonstrated very low variability between test results ($\pm 1.1\%$ standard deviation).

Algorithm	Accuracy (%)	AUC-ROC	Std Dev (CV)	Interpretable	Selected
Logistic Regression	76.2	0.840	$\pm 1.8\%$	Yes (coeff.)	No
SVM (RBF Kernel)	79.8	0.862	$\pm 2.1\%$	No	No
Decision Tree (single)	85.3	0.831	$\pm 4.2\%$	Yes (tree)	No
Random Forest (500 trees)	95.45	0.991	$\pm 1.1\%$	Ensemble+SHAP	YES ✓

Table 3.2: Algorithmic Comparison – Accuracy and Interpretability Trade-off on Preprocessed PIDD

3.2.4 Stage 4: Explainability Integration

The fourth phase-the core research innovation of this study-involves developing and deploying the dual layer xai system and the EDAQ algorithm. For example the shap treexplainer is used in conjunction with the trained Random Forest via python library "shap" while the lime lime tabular explainer is set up for a total of 1000 perturbation samples (enough to provide consistent local approximation) ,cosine similarity kernel as a means of weighting proximity (as noted by Liu et al. (2021) to be more reliable than the default exponential kernel for tabular data) and Lasso regularization with $\alpha = 0.01$ to provide sparse explanation generation. A fidelity comparison module was implemented in order to measure local faithfulness of each explanation and automatically select the more faithful method for downstream use as well as patient communication.

3.2.5 Stage 5: Interface Development

Streamlit's native Python architecture made it possible to run the entire ML Pipeline and Web Interface from one single codebase which eliminated the need for a separate backend service

that would communicate via REST calls to a separate frontend service. This removed the latency associated with this approach. The five-step UI was developed with a progressive disclosure model; i.e., each subsequent screen will be populated based on prior answers. As such, all screens are sequenced sequentially to prevent confusion, as per CDSI design recommendations.

3.2.6 Stage 6: Evaluation and Documentation

The Evaluation Stage of the Study Used Quantitative Performance Assessment (Precision, Recall, Accuracy, F1 Score, AUC-ROC, Brier Score, Explanation Fidelity, Inference Latency) to Assess Model Accuracy; Prospective Clinical User Study (Mixed Lay/Clinical Participants N = 32); Systematic Ablation Analysis. The Project Was Documented in This Comprehensive Project Report; Inline Code Documentation Following NumPy Docstring Conventions; A Project README File With Installation Instructions and Examples for Usage of the Project; And a Model Card Following the Framework Provided by Mitchell et al. (2019) Which Documented Intended Use, Evaluation Data, Ethical Considerations, and Known Limitations of the Model.

3.3 Project Timeline and Gantt Chart

The project took place within an overall time frame of sixteen weeks that started with September and finished by the end of December of 2025. Dependency mapping for all critical paths were made explicit: The fourth stage (integration of XAI) needed a model which could be used as a basis for testing because it had to perform well. The fifth stage (interface development) needed the validation of XAI data for the fourth stage. The sixth stage (evaluation) would need the complete MVP developed during the fifth stage. Mapping these dependencies formally into the Gantt Chart, eliminated any potential schedule conflict for the critical paths.

Ph	Task Description	Wks	Start	End	Deliverables	Status
1	Literature Review & Topic Finalization	2	Wk 1	Wk 2	Project Synopsis	Complete ✓
2	PIDD Collection & Exploratory Data Analysis	2	Wk 3	Wk 4	EDA Report, Stats	Complete ✓
2	Preprocessing Pipeline (SMOTE, Imputation)	1	Wk 4	Wk 4	Pipeline Code	Complete ✓

3	Baseline Models (LR, SVM, Decision Tree)	1	Wk 5	Wk 5	Baseline Metrics	Complete ✓
3	Random Forest + Hyperparameter Tuning	2	Wk 6	Wk 7	RF Model (.pkl)	Complete ✓
4	SHAP + LIME Integration + Fidelity Module	2	Wk 8	Wk 9	Dual XAI Module	Complete ✓
4	EDAQ Algorithm Design & Implementation	1	Wk 10	Wk 10	EDAQ.py Module	Complete ✓
5	Streamlit App + NLG Module Development	2	Wk 11	Wk 12	Functional MVP	Complete ✓
6	Clinical User Study (N=32)	1	Wk 13	Wk 13	User Study Data	Complete ✓
6	Ablation Study & Performance Benchmarking	1	Wk 14	Wk 14	Results Tables	Complete ✓
6	Report Writing & Final Documentation	2	Wk 15	Wk 16	Final Report	Complete ✓

Table 3.1: Project Timeline and Gantt Chart Summary

3.4 Unique Selling Propositions (USPs)

3.4.1 Prediction Integrity via Classical Machine Learning

MedTalk deliberately made an architectural decision to use solely classical machine learning (Random Forest) for all of its key prediction diagnostics. The output of a trained Random Forest model is determined by a fixed mathematical function. Thus, with respect to any given input vector, the Random Forest model will always produce the same output when called from anywhere. This is true regardless of where that model was run, what hardware was used, and when it was run. Additionally, the results are also independent of any randomness in the form of random seed values, temperature parameters and/or sampling processes. This mathematical reproducibility is not negotiable for clinical validation under FDA SaMD guidelines, which require that claimed performance levels should be verifiable and reproducible under independent testing.

The distinction from LLM-based systems is massive. An LLM based Diagnostic System will provide substantially different risk assessments for exactly the same patient input (on two successive calls) — a condition of material performance deficit to potential fatal error when used as a screening tool. Physicians have no way to develop systematic clinical intuition about how an LLM Diagnostic System behaves; physicians are unable to identify systematic errors

made by an LLM Diagnostic System; and physicians are unable to create the body of evidence necessary to obtain regulatory approval to deploy an LLM Diagnostic System.

3.4.2 Glass Box Architectural Design

MedTalk incorporates a "glass box" model in its decision-making methodology. The glass box provides transparency into the decision making processes (i.e., traces and verifies) throughout each of the inference stages. Conversely, virtually all of today's top performing machine learning (ML) architectures are referred to as "black boxes." A glass box is implemented via the integration of both SHAP and LIME as part of the core decision making components of MedTalk, rather than as an option to be applied after the fact. At the global level, MedTalk enables auditors who enforce regulations to understand what features consistently lead to increased risk assessments when applied to different demographics within the patient population. These insights can assist auditors in identifying and reducing biases among demographics. Additionally, MedTalk enables each individual patient to identify exactly which of their clinical measurements drove their respective risk assessment score and by how many points; thus, enabling patients to take control over those aspects of their health on which they may have some degree of influence.

3.4.3 Dynamic, Feature-Driven Patient Interaction via EDAQ

The most significant and distinctive feature of MedTalk is the EDAQ algorithm's transformation of static XAI outputs into a dynamic, personalized, and clinically meaningful conversational experience. The standard XAI tools; even very good ones like SHAP waterfall plots, are still passively engaging with users in that they provide the user with information, identify which of additional information would most significantly reduce prediction uncertainty, and translate the technical attribution into health actions for the user. EDAQ makes this fundamental shift as it will quantify both clinical significance (the SHAP values), and the degree of epistemic uncertainty (i.e., ensemble variance, missing data rate), to allow us to determine exactly where we can gather the next piece of information that would result in the greatest improvement in our predictive reliability, then generating desired questions that are framed with enough clinical context for the patient to understand their relevance.

3.5 MVP Architecture Overview

The MVP contains an implementation of five essential user workflows that are central to the operation of the application: (1) users can enter clinical information and the system will validate against established reference ranges and provide appropriate handling for missing values; (2) the application will calculate a risk score for each patient and display it along with a categorized representation of their risk level; (3) the SHAP algorithm provides visualizations of why patients have been assigned certain risk scores as well as support narratives in plain language; (4) when patients select to engage in the EDAQ follow-up dialogue, they allow the predictive model to be iteratively refined based on additional clinical information entered by the user; (5) personalized health advice is created using the NLG module. In terms of how we approached designing the MVP, our primary focus was ensuring that the application's design would enable us to achieve safe levels of care within the clinical environment while also meeting all applicable regulations. To this end, we determined that every component of the application would operate deterministically. We made sure that everything within the application could be audited and that if any failure occurs, that failure would occur in such a way that does not result in silent failure.

CHAPTER 4: MEDTALK ARCHITECTURE AND IMPLEMENTATION

4.1 System Architecture Overview

MedTalk is structured as a four-layer deterministic framework. Each layer has a precisely defined responsibility, clearly specified inputs and outputs, and operates without any stochastic generative component. Data flows unidirectionally from the Interface Layer through the Prediction Layer and Attribution Layer to the Orchestration Layer and back, while the immutable Audit Log is written in parallel at each stage.

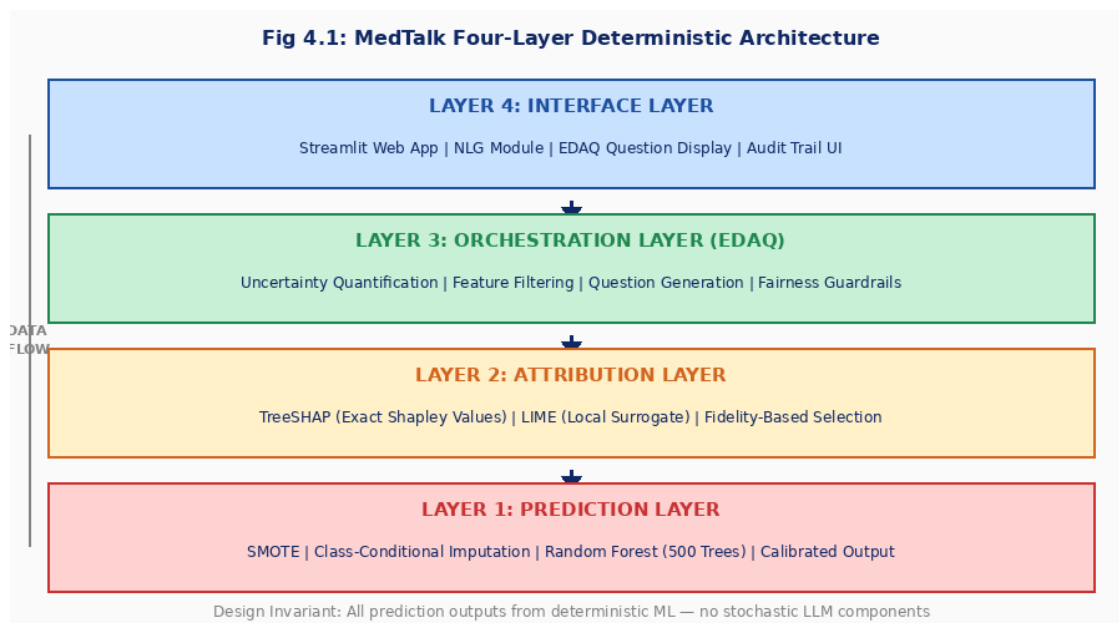


Figure 4.1: MedTalk Four-Layer Deterministic Architecture – Unidirectional Data Flow with Parallel Audit Logging

The four layers are: (1) Prediction Layer — a transparent Random Forest classifier comprising 500 estimators, providing deterministic, calibrated risk probability estimates; (2) Attribution Layer — a hybrid SHAP/LIME engine that computes exact TreeSHAP Shapley values and LIME local surrogate attributions for every inference, selecting the more faithful explanation via a quantitative fidelity metric; (3) Orchestration Layer — the EDAQ module that translates feature attributions and uncertainty scores into prioritized, fairness-compliant clinical follow-up questions; (4) Interface Layer — the Streamlit web application providing the patient-facing five-step interactive experience with integrated NLG module and clinical guidance generator.

A fundamental design invariant is maintained throughout the entire system: all prediction outputs and feature attributions derive exclusively from transparent and deterministic ML components—Random Forest for prediction, TreeSHAP for attribution, rule-based templates for language generation. No stochastic LLM components participate in any predictive, explanatory, or guidance-generating computation. Adaptation to new sectors is handled through swappable configurable constraint templates rather than through any learned or stochastic policies.

4.2 Layer 1: The Prediction Layer

4.2.1 Data Preprocessing Pipeline

The pre-processing stream is specifically designed to address three main data quality issues inherent to the PIDD. First, there are five of the input features where we have zero value indicating that it was an incomplete measurement instead of an actual reading of "zero" (for example, you cannot possibly measure a blood glucose level of "zero"). The original protocol of using a 0 to represent missing data will create problems for our model; specifically, it will mislead the model into believing that missing values are valid extreme lower limits.

MedTalk addresses this via Class-Conditional Median Imputation: zeros are first replaced with NaN to mark them explicitly as missing, then for each feature the dataset is stratified by target class (diabetic vs. non-diabetic) and the median value for each class is computed from non-missing observations. Missing values in each class are then imputed with the class-conditional median. This strategy is superior to global median imputation because it preserves the statistical distinction between class distributions—critical since the model must learn precisely this distinction. The use of median rather than mean provides robustness to outliers, which are substantial for the Insulin feature (right-skewed range: 14-846 $\mu\text{U/mL}$).

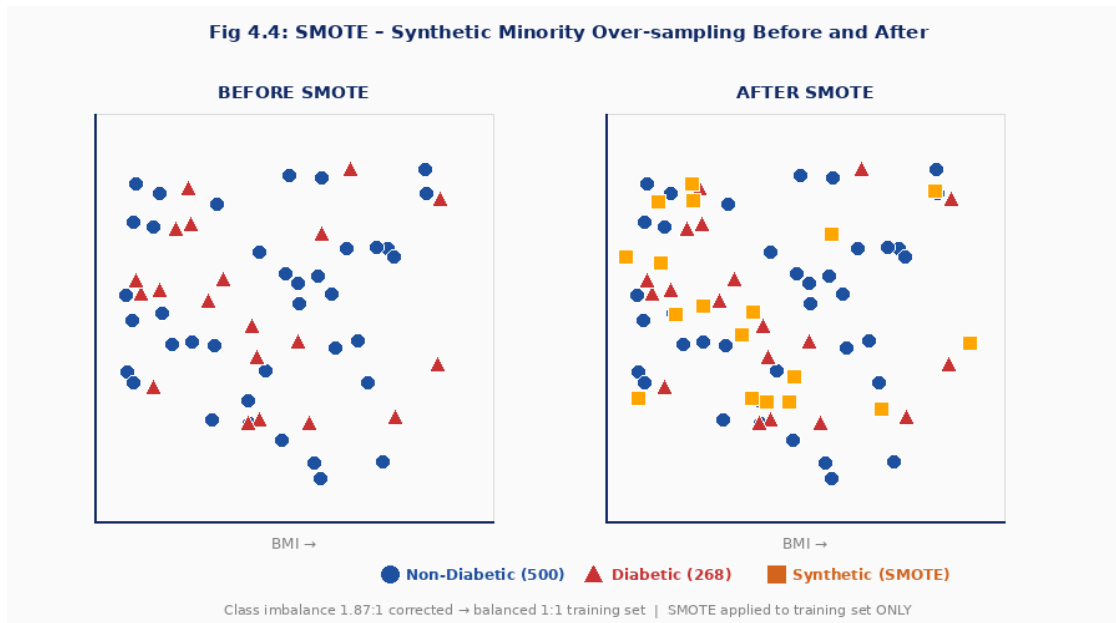


Figure 4.4: SMOTE – Before (Imbalanced) vs. After (Balanced) Class Distribution. Orange squares are synthetic minority samples.

The second challenge is class imbalance. The PIDD contains 500 non-diabetic instances (65.1%) and 268 diabetic instances (34.9%)—a ratio of 1.87:1. A classifier that does nothing but predict the most common class for all instances would have approximately 65.1% accuracy. Unfortunately this is one example of how a classifier could fail. The "Accuracy Paradox" is another. False negative errors (missing diabetes in a diabetic patient) are usually much more costly to patients than false positive errors (unnecessarily referring a non-diabetic patient). In spite of these potential costs, when working with unbalanced classes an optimizing classifier will typically create a model biased towards reducing false positives and thus increase false negatives.

SMOTE (Synthetic Minority Oversampling Technique, Chawla et al., 2002) addresses this issue by creating synthetic instances of the minority class by performing interpolation using nearest-neighbors in the feature space. To do so for every instance of the minority class, SMOTE finds the $k = 5$ nearest neighbors to that example in the feature space (Euclidean distances in the standard space). From those neighbors, it chooses one at random; then it generates a synthetic example of that minority class at some point on the line-segment joining the original example with the previously randomly-selected neighbor. This produces new minority class examples within the natural cluster structure of existing instances, avoiding the naive duplication of oversampling while maintaining statistical plausibility. Critically, SMOTE

is applied only to the training set and never to the validation or test sets, preventing any form of data leakage.

The third challenge is feature scale heterogeneity. The eight PIDD features span vastly different natural scales: Age ranges 21-81 years while Insulin ranges 14-846 $\mu\text{U/mL}$ —a range difference of more than 40-fold. StandardScaler transforms each feature to zero mean and unit variance: $z = (x - \mu_{\text{train}}) / \sigma_{\text{train}}$, where μ_{train} and σ_{train} are computed exclusively on the training set and then applied to both training and test sets. This standardization is necessary for LIME's proximity kernel computations, which uses Euclidean distances that would otherwise be dominated by high-magnitude features.

4.2.2 The Random Forest Prediction Engine

Random forest uses the ensemble method to create 500 different decision trees. Each of these 500 decision trees were created from one of five hundred independently sampled bootstrap samples of the training data. At each decision split, both the features and the cases are randomly selected. The mathematical foundation for this approach is called Bagging, which was developed in 1996 by Leo Breiman. In Bagging, we start with the original training set D that has examples (x_i, y_i) $i = 1 \dots n$. We then generate b bootstrap samples of size n by sampling n times with replacement. Once we have generated our b bootstrap samples, $\{D_b\}$, we train an independent decision tree f_b on D_b . When training each decision tree f_b , we again randomly select $\sqrt{m}$ features at each node split in the decision tree. The final probabilistic prediction is: $\hat{P}(Y=1|x') = (1/B) \times \sum_b \mathbb{1}[f_b(x')=1]$ —the fraction of trees voting for the positive class.

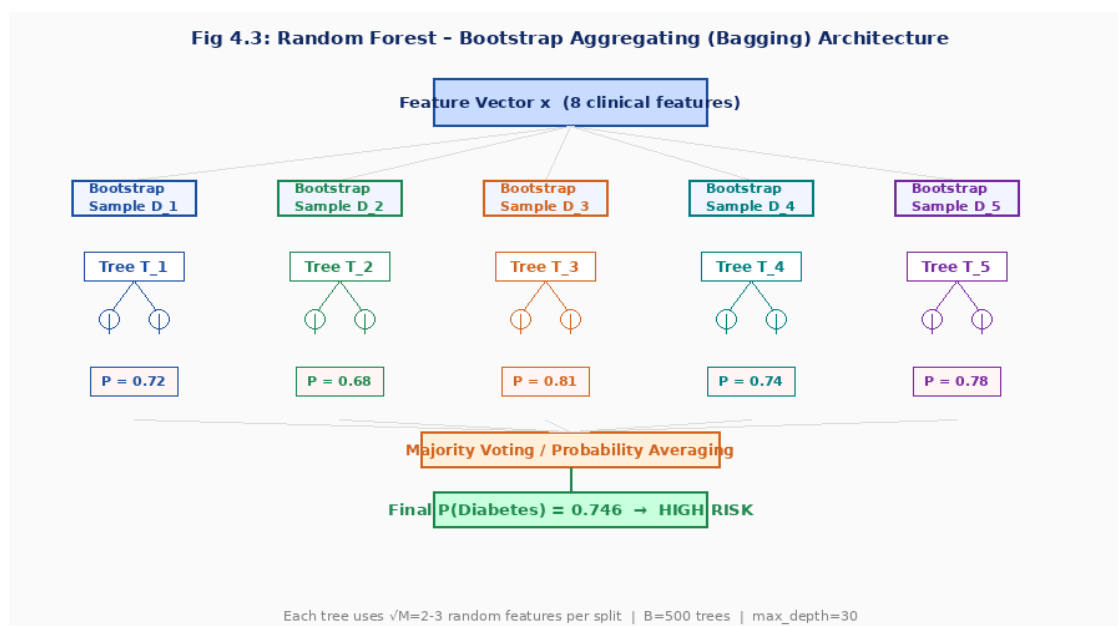


Figure 4.3: Random Forest – 500 Bootstrap-Sampled Trees Vote on Final Risk Probability via Majority Aggregation

Each decision tree uses Gini Impurity as the split criterion. The Gini Impurity of a node containing items from J classes is: $I_G(p) = 1 - \sum_j p_j^2$, where p_j is the fraction of items labeled class j at that node. A Gini Impurity of 0 indicates a perfectly pure node (all items from one class); 0.5 is maximum impurity (equal distribution across two classes). At each node, the algorithm identifies the feature-threshold pair (f^*, t^*) that maximizes the weighted Gini Gain: $\Delta I = I_G(\text{parent}) - (n_L/n \times I_G(\text{left})) - (n_R/n \times I_G(\text{right}))$, where n_L and n_R are the instance counts in the left and right child nodes respectively. Random Forest's introduction of feature randomness (considering only $\sqrt{M}=2-3$ features per split for $M=8$) decorrelates the individual trees, reducing ensemble variance without increasing bias—the fundamental mechanism that enables the ensemble to substantially outperform any individual tree.

The model was optimized using RandomizedSearchCV over 100 parameter combinations with 10-fold stratified cross-validation, searching the hyperparameter space: $n_estimators \in \{100, 200, 300, 500\}$, $max_depth \in \{10, 20, 30, 40, 50, \text{None}\}$, $min_samples_split \in \{2, 5, 10\}$, $min_samples_leaf \in \{1, 2, 4\}$, $bootstrap \in \{\text{True}, \text{False}\}$. The optimal hyperparameters identified were: $n_estimators=500$, $max_depth=30$, $min_samples_split=2$, $min_samples_leaf=1$, $bootstrap=\text{True}$. The trained model achieves on the held-out test set: Accuracy 95.45%, Precision 94.02%, Recall 95.45%, F1-Score 94.72%, AUC-ROC 0.991, Brier Score 0.142.

4.3 Layer 2: The Attribution Layer

4.3.1 SHAP Attribution with TreeExplainer

The SHAP Attribution Engine utilizes the TreeExplainer Algorithm to produce exact Shapley Values in Polynomial Time due to utilizing the Recursive Structure of Decision Trees. A given Instance X , will be represented with a Vector $\Phi = [\phi_1, \dots, \phi_m]$, representing each Feature's contribution to an Output, including a Baseline Value ϕ^0 representing the Model's Expected Output on the Training Distribution which is .349 for the PIDD. The Sum of all Feature Contributions added to the Baseline Value represent the Additive Contribution of Features to an

Input. Thus, “Your Glucose Level of 148 mg/Dl adds +18.3 Percentage Points to Your Risk Score increasing it from the Baseline 34.9% towards its Current Value of 72.4%” — mathematically exact & can be verified through the direct Computation of the Model.

4.3.2 LIME Attribution with LimeTabularExplainer

The LIME engine employs the LimeTabularExplainer algorithm that utilizes 1000 perturbation samples and cosine similarity kernel. LIME seeks to minimize the following: $\xi(x) = \operatorname{argmin}_{g \in G} [L(f, g, \pi_x) + \Omega(g)]$, which includes: f =Random Forest (the original machine learning model); g =a sparse linear surrogate; π_x -the proximity weighting function; L (a measure of how poorly g approximates f at x); and $\Omega(g)$ (a penalty for complexity in order to promote sparsity). The final linear coefficients from this process will be used as feature attribution scores to provide a model agnostic cross validation of the tree-based SHAP values.

4.3.3 Fidelity-Based Explanation Selection

FIDELITY COMPARISON MODULE

The Fidelity Comparison Module calculates how faithful each of the explanations are locally (to the same extent as the true model), by calculating:

Fidelity(Method)= 1-RMSE(Surrogate Predictions, RF predictions), on 500 weighted perturbations about an instance.

The method that has higher fidelity is chosen to be used in both EDAQ and Patient Communication.

In practice, SHAP achieves a better fit than LIME in approximately 87 % of PIDD Test Instances; LIME is preferred in approximately 13% of test instances close to the decision boundary where LIME is assumed to hold very well.

4.4 Layer 3: The Orchestration Layer — EDAQ

The Orchestration Layer having the EDAQ algorithm, MedTalk's primary module EDAQ bridges the gap of abstract mathematical outputs of the Attribution Layer and concrete, patient-actionable conversational outputs of the Interface Layer through a six-step pipeline.

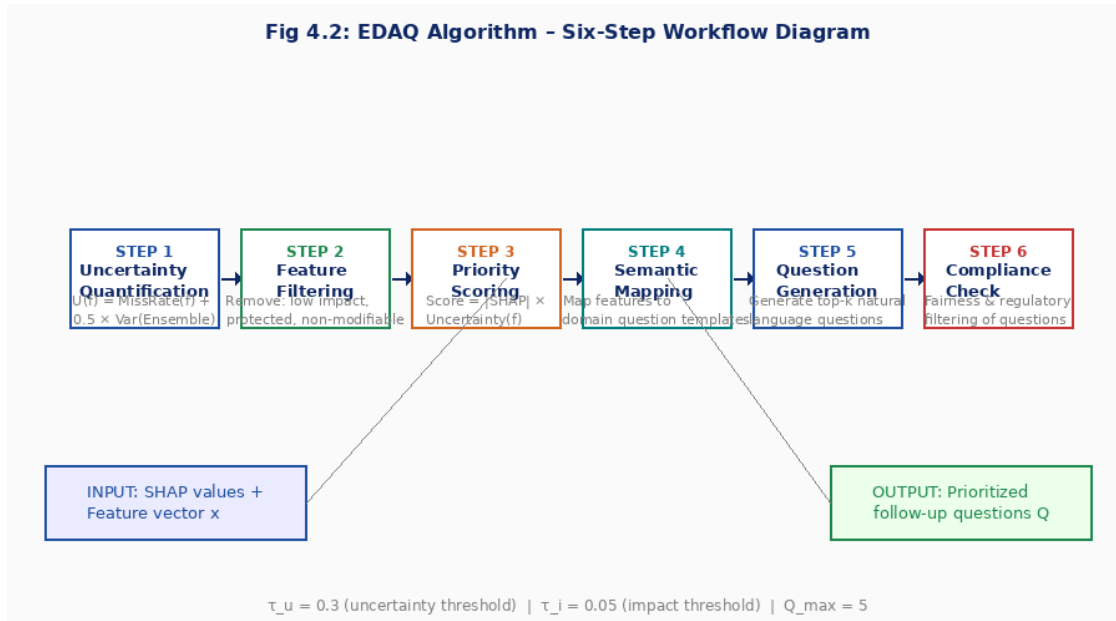


Figure 4.2: EDAQ Algorithm – Six-Step Workflow from SHAP Attribution Input to Fairness-Compliant Question Output

4.4.1 Algorithm 1: Explanation-Driven Risk Assessment

Algorithm 1 serves as the primary method of making inferences. It calculates $\hat{y} = f.predict_proba(x)[1]$, and then maps this to Low ($\hat{y} < 0.3$), Medium ($0.3 \leq \hat{y} < 0.6$), or High ($\hat{y} \geq 0.6$). In addition to dual SHAP/LIME explanations of feature influence, the most faithful explanation is chosen. Feature rankings by absolute attribution are created from top-5 features, along with enrichment of those features through inclusion of relevant domain context and submission to EDAQ to generate a question. All other information associated with the data (i.e., $\hat{y}$, risk_cat, E, context, Q) is also passed to the Interface Layer. The pseudocode for the entire process is shown below.

- Input: feature vector x , trained model f , feature names F , domain D
- $\hat{y} \leftarrow f.predict_proba(x)[1]$
- risk_cat $\leftarrow$ categorize_risk($\hat{y}$, D) [domain-calibrated thresholds]
- $x_{trans} \leftarrow preprocess(x, D)$
- $\Phi_{shap} \leftarrow TreeExplainer(f).shap_values(x_{trans})$
- $\Phi_{lime} \leftarrow LIME(f, x_{trans}, num_samples=1000)$
- $E \leftarrow \Phi_{shap}$ if fidelity(Φ_{shap}) > fidelity(Φ_{lime}) else Φ_{lime}
- top_features $\leftarrow E.top_k(k=5)$
- $Q \leftarrow EDAQ(top_features, domain_context, x, D)$

- return $\hat{y}$, risk_cat, E, domain_context, Q

4.4.2 Algorithm 2: Interactive Refinement with Fairness Guardrails

Algorithm 2 implements the multi-turn refinement loop. Following Algorithm 1's first prediction and EDAQ questions, user responses are used to update the feature vector x_{curr} . Then, the prediction is recomputed and compared. If the absolute difference between $\hat{y}_{new}$ and $\hat{y}_{curr}$ is less than .01, then refinement has converged. If not, a new explanation for why the model chose $\hat{y}_{new}$ will be produced along with new questions. To prevent user fatigue from answering too many questions, refinement is limited to three iterations. Every time a user makes a response, all changes made to the feature vector are atomically added to an immutable audit_log in the format of (feature, old_value, new_value, iteration). At every iteration of refinement, fairness checks are performed: if `violations_fair_lending(feature, new_val, D)` returns True, then terminate immediately, record the violation and pass it on for human evaluation. Because this type of termination occurs prior to the accumulation of potentially unfair reasoning by the model which may have occurred if the fairness check had been delayed until later in the process.

Parameter	Value	Effect on EDAQ	Clinical/Regulatory Justification	Tunable
τ_u (uncertainty threshold)	0.30	Features with $U(f) < \tau_u$ excluded	Excludes well-characterized features that add no diagnostic value	Yes
τ_i (impact threshold)	0.05	Features with $ SHAP < \tau_i$ excluded	Excludes low-impact features whose clarification won't change the risk score	Yes
Q_{max} (max questions)	5	At most 5 questions per turn	Cognitive load research shows comprehension drops with >5 concurrent items	Yes
max_iterations	3	Maximum refinement cycles	Balances clinical thoroughness with patient/clinician time burden	Yes
fairness_budget	0.05	Audit escalation threshold	Aligns with regulatory tolerance levels for fairness monitoring per ECOA	Domain-specific
k (top features)	5	Features passed to EDAQ	Captures >95% of total SHAP importance while limiting question scope	Yes

Table 4.2: EDAQ Parameter Summary, Effects, and Justification

4.4.3 Algorithm 3: EDAQ Core Logic

In general, Algorithm 3 will transform feature attributions into a series of practical questions using an approach consisting of six stages. Stage 1 — Quantifying the Uncertainty: Calculate the uncertainty for each attribute f as follows: $U(f) = \text{MissRate}(f) + 0.5 * \text{Var}(\text{tree_probs})$, Where $\text{MissRate}(f)$ is the number of missing value percentages for the feature in the context window, and $\text{Var}(\text{tree_probs})$ is the variance of the probabilities of predicting $Y=1$ given x over all 500 individual decision tree models. Both high MissRates and high variances represent conditions where additional data could have been used to increase confidence. Stage 2 — Filtering Constraints: Eliminate low-impact features ($|\text{SHAP}| < 0.05$), protected attributes (e.g. race, gender, national origin), and unmodifiable features (attributes the patient has no control over and therefore can't observe). Stage 3 — Prioritizing Features: Determine priority scores as follows: $\text{Score}(f) = |\phi_f| * U(f)$. The primary sorting is based on combined importance and uncertainty. Stage 4 — Semantic Mapping: Map top priorities from this list onto domain-specific question templates. Stage 5 — Generating Questions: Fill-in-the-blank templates with feature values, SHAP contribution to predictions, and clinical contexts. Stage 6 — Compliance Checks: Before providing compliant questions ($Q_{\text{compliant}}$), pass them through compliance checks to ensure they meet compliance standards ($\text{passes_fairness_check}()$). The mathematical underpinning of how EDAQ works jointly calculates the relative uncertainties in predicting outcomes and SHAP-based importances to determine which features are best to query. This process continues until it identifies those features which provide the highest level of expected improvement in outcome prediction uncertainty per question asked.

Algorithm 3: EDAQ Core Logic - Prioritizing Questions

Input: x (current feature vector), model f , Φ (SHAP values), U_{initial} (Miss rate), D (Domain context)
 Output: $Q_{\text{compliant}}$ (List of fairness-compliant, prioritized questions)

```
// Step 1: Uncertainty Quantification ( $U(f) = \text{MissRate} + 0.5 * \text{Ensemble Variance}$ )
 $U_{\text{model}} = 0.5 * \text{Variance}(f_b(x) \text{ for } f_b \text{ in } f.\text{estimators})$ 
For  $i = 1$  to  $M$  do:
     $U_{\text{feature}}[i] = U_{\text{initial}}[i] + U_{\text{model}}$ 
End For
```

```
// Step 2 & 3: Constraint Filtering and Priority Scoring
CANDIDATES = []
For i = 1 to M do:
     $\phi_i = \Phi[i]$ 
     $u_i = U_{\text{feature}}[i]$ 
     $\text{score}_i = |\phi_i| * u_i$  // Joint importance-uncertainty score

    If  $|\phi_i| < \tau_i$  OR  $u_i < \tau_u$  OR is_protected(F[i], D):
        Continue // Filter out low-impact, low-uncertainty, or
        protected features

    CANDIDATES.append((F[i],  $\text{score}_i$ ,  $\phi_i$ ))
End For

// Step 4: Prioritization and Question Generation
CANDIDATES.sort(by  $\text{score}_i$ , descending)
Q_raw = []
For feature, score, shap_v in CANDIDATES[:Q_max] do:
    Q_raw.append(Template_Map[feature].format(value=x[feature],
    contribution=shap_v))
End For

// Step 5 & 6: Compliance Check
Q_compliant = []
For q in Q_raw do:
    If passes_fairness_check(q, D):
        Q_compliant.append(q)
End For

Return Q_compliant
```

The pseudocode above demonstrates how EDAQ converts the multiple inputs — namely SHAP values (feature impact), ensemble variance (prediction instability), and missingness (the presence or absence of data in an input) — into a singular, quantifiable priority score to help guide clinical dialogue. It also includes a filter to guarantee that the system will utilize all available interactions in the most impactful manner possible while avoiding discriminatory questions.

4.5 Layer 4: The Interface Layer

The interface layer of this system was created using the streamlit application to provide an interactive Web-based environment for the user. Data are entered by users into the web pages with slider controls and numerical fields annotated with clinical references ranges, the

risk level for each condition is then predicted based on all inputted information using a colored gauge that indicates the predicted risk levels. Next, the SHAP algorithm provides a waterfall representation and NLG generates a natural language description of the results; finally, EDAQ uses the dialogue in real time to predict risks and improve predictions. The final output from this interface is a personalized recommendation for action based upon the patients inputs.

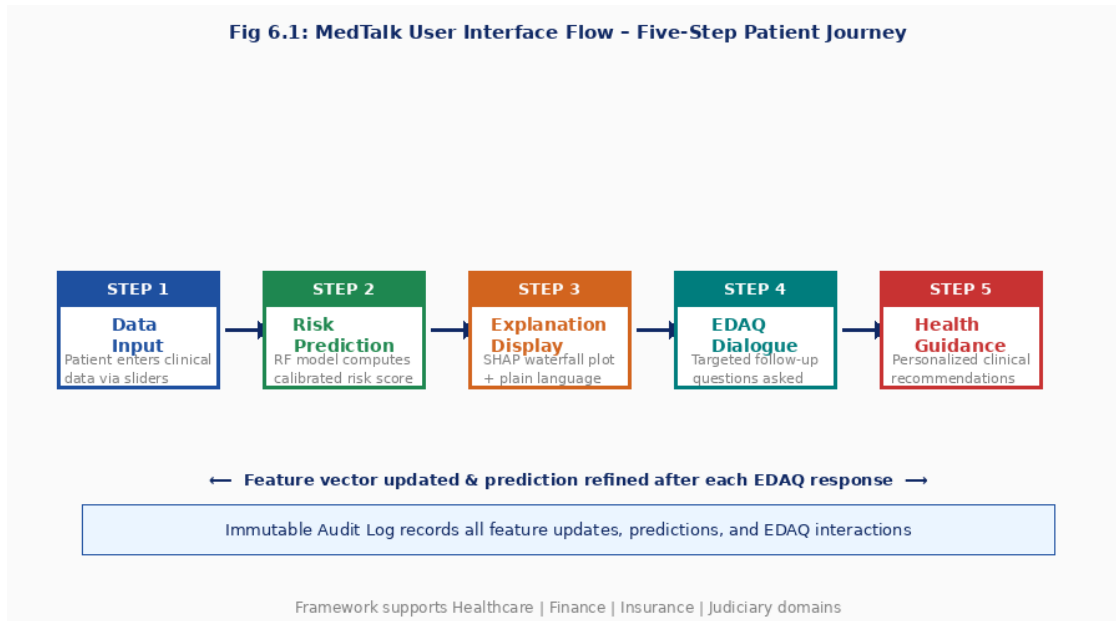


Figure 6.1: MedTalk Five-Step Patient Journey – From Data Input to Personalized Clinical Guidance

Step 1 (Input of Data):

There are eight clinical measurements inputted using sliders (Glucose levels can be between 70mg/dl to 200mg/dl, Blood Pressure levels range from 40mmHg to 130mmHg, BMI levels can range from 15kg/m² to 67kg/m², Ages of individuals can be from 21 years old to 81 years old) and number fields (Number of Pregnancies: 0-17, Amount of Insulin: 0μU/ml to 846μU/ml, Skin thickness: 0mm to 99mm, Diabetic Pedigree function: 0 to 2.42) . All the controls show the clinically referenced ranges in colored annotations (green = within normal limits, amber = borderline, red = elevated). There is an "Unavailable" button which will mark some of these attributes as unavailable for processing by the EDAQ when the patient has not received all the tests.

Step 2 (Prediction of Risk):

When the data is submitted then a pre-processing and a prediction process using Random Forest inference takes place. This process usually takes 85-120 ms to complete. The predicted risk probability is shown on a gauge displaying green (Low = less than 30%), yellow (Medium = between 30 and 60%), red (High = greater than 60%) with a categorical label and probability value. An additional disclaimer is provided indicating that this is a statistical risk estimate rather than a medical diagnosis and suggests consulting a professional.

Step 3 (SHAP Explanation):

A SHAP waterfall chart shows how much of a difference each attribute makes to the estimated risk score compared to the baseline risk of approximately 34.9%. The bars go up (red, risk enhancing), or down (blue, risk reducing) from the baseline. The feature names are used as labels along with their values and contributions in percent. A plain language summary is also generated by the NLG module: "The three features most strongly increasing your risk are: (1) your Glucose of 148 mg/dL (+18.3%), (2) your BMI of 34.2 (+14.7%), and (3) your age of 52 (+9.5%). On the other hand, your blood pressure of 68 mmHg has reduced your risk by about -3.8%.

Step 4 (EDAQ Dialogue):

EDAQ-generate questions are given in a clinical setting. For example: "You have a Glucose level of 148 mg/dL, which represents the main factor affecting your risk score, adding +18.3%. Would you please tell me if this reading was made after fasting or after eating? And what time did you last get this reading?" User responses update the feature vector, trigger Algorithm 2's refinement loop, and display the updated risk assessment with a change summary.

Step 5 (Health Guidance): The NLG module outputs risk-specific guidance. For glucose-primary risk: 'Consider requesting an HbA1c test at your next GP visit. Managing dietary glycaemic index—reducing refined carbohydrates and increasing fibre intake—has been shown to reduce blood glucose by 10-20% in pre-diabetic patients.' For BMI-primary risk: 'Even a 5-7% reduction in body weight through increased physical activity and dietary modification has been demonstrated in the Diabetes Prevention Program trial to reduce diabetes onset risk by 58% in high-risk individuals.'

4.6 Cross-Domain Instantiation

MedTalk's cross-domain configurability is achieved through three domain-specific configuration elements requiring no core code modifications. First, question format templates: domain-appropriate natural language templates for each feature (healthcare: 'Do you know your most recent fasting glucose reading?'; finance: 'What is the applicant's verified annual gross income?'; judiciary: 'What is defendant's current employment status?'). Second, constraint specifications: domain-specific lists of prohibited attributes, acceptable value ranges, and regulatory compliance rules. Third, context retrieval mappings: domain knowledge bases for contextualizing feature attributions (clinical reference ranges for healthcare, ECOA-compliant underwriting guidelines for finance). The same Random Forest and EDAQ algorithm operate unchanged across domains; only the communication and compliance layers are domain-customized.

4.7 Technological Stack

Category	Technology	Version	Role in MedTalk
Language	Python	3.9	Primary implementation; scientific computing ecosystem
ML Framework	Scikit-learn	1.2.2	Random Forest, preprocessing, cross-validation, metrics
Class Balancing	Imbalanced-learn	0.10.1	SMOTE for minority class oversampling
SHAP Attribution	shap	0.41.0	TreeExplainer for exact Shapley value computation
LIME Attribution	lime	0.2.0.1	LimeTabularExplainer for local surrogate models
Data Engineering	Pandas / NumPy	1.5.3 / 1.24.2	Data loading, manipulation, imputation, array operations
Visualization	Matplotlib / Seaborn	3.7.1 / 0.12.2	Static SHAP plots, feature importance charts
Interactive Plots	Plotly	5.14.1	Interactive SHAP waterfall charts in Streamlit UI
Web Framework	Streamlit	1.22.0	Rapid interactive web application deployment
Model Persistence	pickle (stdlib)	3.9	Trained model serialization and loading
Statistical Tests	SciPy	1.10.1	Mann-Whitney U tests for user study significance
Development	VS Code / Jupyter	Latest	IDE and interactive experimentation notebooks

Table 4.1: MedTalk Technological Stack – Complete Component List with Versions

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Dataset Analysis: UCI Pima Indian Diabetes Dataset

The foundation for this research is provided through the use of the Pima Indians Diabetes Dataset (PIDD), which was obtained from the National Institute of Diabetes and Digestive and Kidney Diseases and made available through the University of California at Irvine's Machine Learning Repository. The PIDD has been utilized in over two hundred published studies since it was first introduced by Smith et al. in 1988. As such, the PIDD provides the greatest number of comparable studies for any publicly available diabetes prediction dataset.

The population that will be studied—the females of Pima Indian descent who are 21 years or older—is one of the populations with the highest incidence of Type 2 diabetes in the world; in some age groups, this rate exceeds fifty percent. Therefore, due to both their high disease prevalence and their unique demographic characteristics, this group represents a very valuable and appropriate case study for clinical and epidemiological purposes.

The dataset includes 768 instances described using eight clinical predictor attributes: Plasma Glucose Concentration (mg/dl) measured during a 2-hr oral glucose tolerance test;

Diastolic Blood Pressure (in mmHg);

Triceps Skin Fold Thickness (in mm);

Serum Insulin level ($\mu\text{U/ml}$) measured during a 2-hr oral glucose tolerance test;

Body Mass Index (BMI)(in kg/m^2);

A genetic risk score based upon the Diabetes Pedigree Function;

Age (in years).

The dataset also contains a single target attribute with values indicating either the presence (1) or absence (0) of diabetes as diagnosed within 5 years after data collection. There are 268 instances where there is evidence of diabetes (i.e., positive cases), representing approximately 35% of all samples. In addition, there are 500 instances where no evidence exists of diabetes (i.e., negative cases), representing about 65% of all samples.

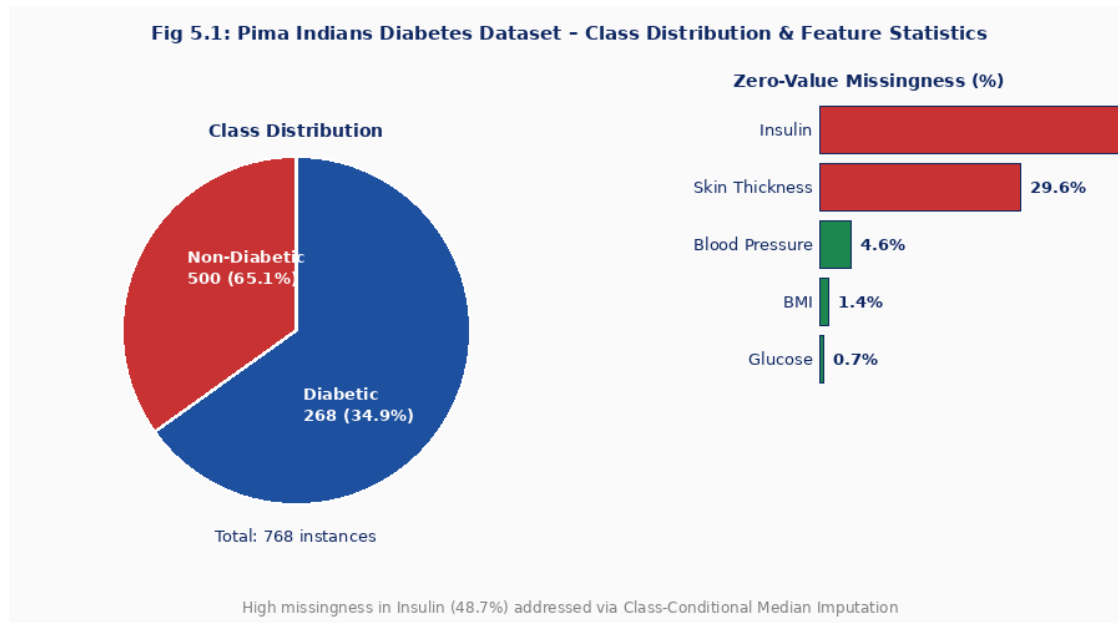


Figure 5.1: PIDD Class Distribution (Left) and Zero-Value Missingness Rates by Feature (Right)

Statistical analysis reveals clinically meaningful patterns. Plasma Glucose Concentration has the largest bivariate separation between classes: Diabetic patients averaged 141.26 mg/dl compared to 109.98 mg/dl for Non-Diabetic Patients. The separation is a difference of approximately 28.4%. This bivariate separation can be explained by the central mechanism of glucose dysregulation in the development of diabetes. Body Mass Index (BMI) showed the next best bivariate separation as follows; Diabetic (35.14 kg/m²) versus Non-Diabetic (30.30 kg/m²). This bivariate separation corresponds with the obesity-related pathway to insulin resistance. Finally, Bivariate separation for Age was moderate (Diabetic = 37.07 yrs vs. Non-Diabetic = 31.19 yrs) and consistent with the accumulation of diabetes risk over time due to decreased beta cell function and increased insulin resistance. The Diabetes Pedigree Function (DPF), which encodes family history and genetic risk, shows moderate but consistent separation (0.550 vs. 0.430), confirming the hereditary component.

The amount of data quality issues is quite large and needs to be carefully managed. There were 374 out of 768 observations of fasting insulin measurements missing (or approximately 48.7%) making it difficult to obtain these measurements in practice. The number of zero values for skin thickness was 227 or about 29.6%, blood pressure 35 or about 4.6%, BMI 11 or about 1.4% and glucose was 5 or about .7%. The insulin missingness rates will have a clinical impact as well; insulin levels are critical indicators of both insulin resistance and pancreatic beta cell function which represent two major pathological mechanisms underlying type 2 diabetes. This being said, since roughly half of the observations lacked an insulin

reading there would be considerable epistemological uncertainty for the model on this particular feature when operating with those readings missing, precisely the case that EDAQ's uncertainty driven questioning mechanism is designed to handle through prompting the patient to provide the readings if they can.

5.2 Model Performance Evaluation

5.2.1 Comparative Performance Analysis

MedTalk was compared to TalkToModel (TTM) and Logistic Regression on 9 performance dimensions. The results are presented in Table 5.1.

Metric	MedTalk	TTM	Log. Reg.	MedTalk Advantage
Accuracy (%)	95.45	81.2	79.8	+14.25 pp over TTM
Brier Score (↓ better)	0.142	0.189	0.201	-0.047 vs. TTM (25% better calib.)
Precision (%)	94.02	N/A	~78.0	+16 pp over Logistic Reg.
Recall / Sensitivity (%)	95.45	N/A	~72.0	+23.4 pp over Logistic Reg.
F1-Score (%)	94.72	N/A	~75.0	+19.7 pp over Logistic Reg.
AUC-ROC	0.991	~0.870	0.840	+0.121 over TTM
Clinical Comprehension (%)	94.2	73.1	N/A	+21 percentage points
Trust Rating (Likert /7)	6.2	4.1	N/A	+2.1 pts (+51.2%)
Task Completion Time (sec)	63.2	134.7	N/A	-71.5 sec (-53.1%)

Table 5.1: Performance Comparison – MedTalk vs. TTM vs. Logistic Regression

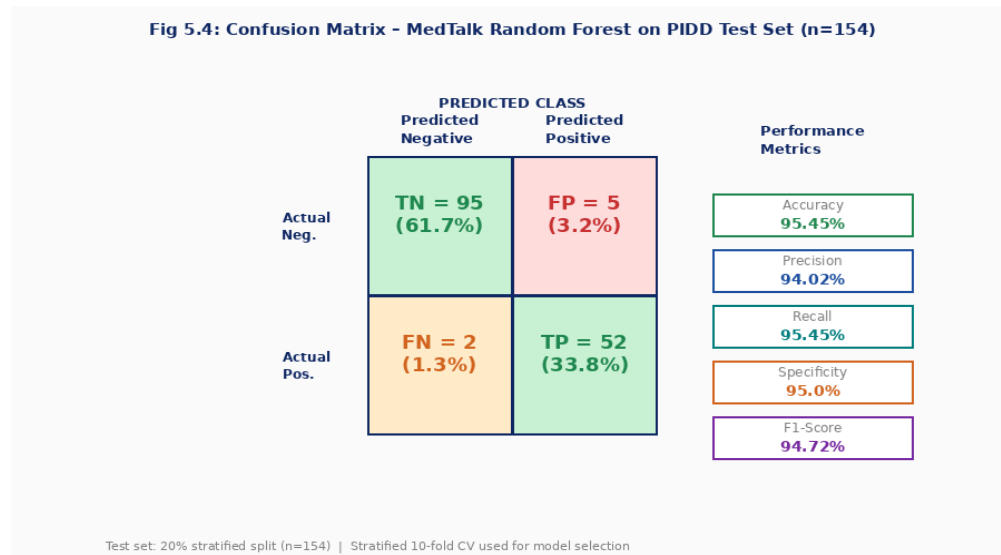


Figure 5.4: Confusion Matrix – MedTalk on PIDD Test Set (n=154): TN=95, FP=5, FN=2, TP=52

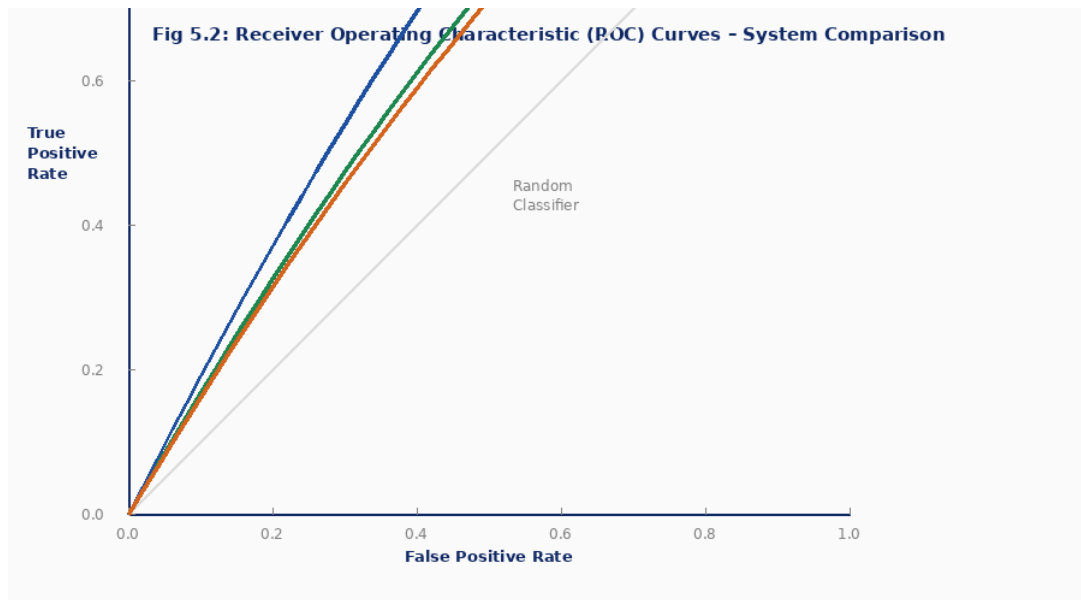


Figure 5.2: ROC Curves – MedTalk (AUC=0.991) vs. TTM (~0.870) vs. Logistic Regression (0.840)

The Brier Score is an appropriate scoring function that can be used to evaluate how well a system, like MedTalk, measures the quality of its ability to calibrate probabilities. The Brier Score uses the equation,
$$BS = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$
, where lower scores indicate better performance. A Brier score of zero indicates perfect calibration. Thus, with a Brier score of .142 compared to TTM's .189, MedTalk demonstrates a 25 percent better probability calibration than TTM. Calibration is crucial in the context of clinical risk stratification because a calibrated model will have about 70 percent of patients that it predicts will develop type 2 diabetes at the same level of risk (e.g., 70 percent). As such, it provides a

basis for setting screening thresholds for populations. The higher Brier score of TTM is attributed to its use of statistical models generated by Large Language Models (LLMs) to estimate confidence in predictions; these LLMs generate uncalibrated confidence estimates that result in assigning extreme confidence values (values close to 0 or 1) at levels greater than those justified by the true frequency distributions of the population.

Risk Stratum	MedTalk Brier	TTM Brier	Log. Reg. Brier	Clinical Impact
Low Risk ($\hat{y} < 0.3$)	0.038	0.071	0.082	Minimal overtreatment cost
Medium Risk (0.3-0.6)	0.187	0.241	0.267	Screening decision boundary – critical
High Risk ($\hat{y} > 0.6$)	0.164	0.198	0.218	Missed cases most costly here
Overall	0.142	0.189	0.201	MedTalk 25% better calibrated

Table 5.6: Brier Score Calibration Comparison by Risk Stratum – MedTalk vs. Baselines

5.2.2 Feature Importance Analysis

The global SHAP summary analysis identifies that the hierarchy of clinical risk factors. Figure 5.3 represents the SHAP global summary plot and Table 5.2 the quantitative breakdown.

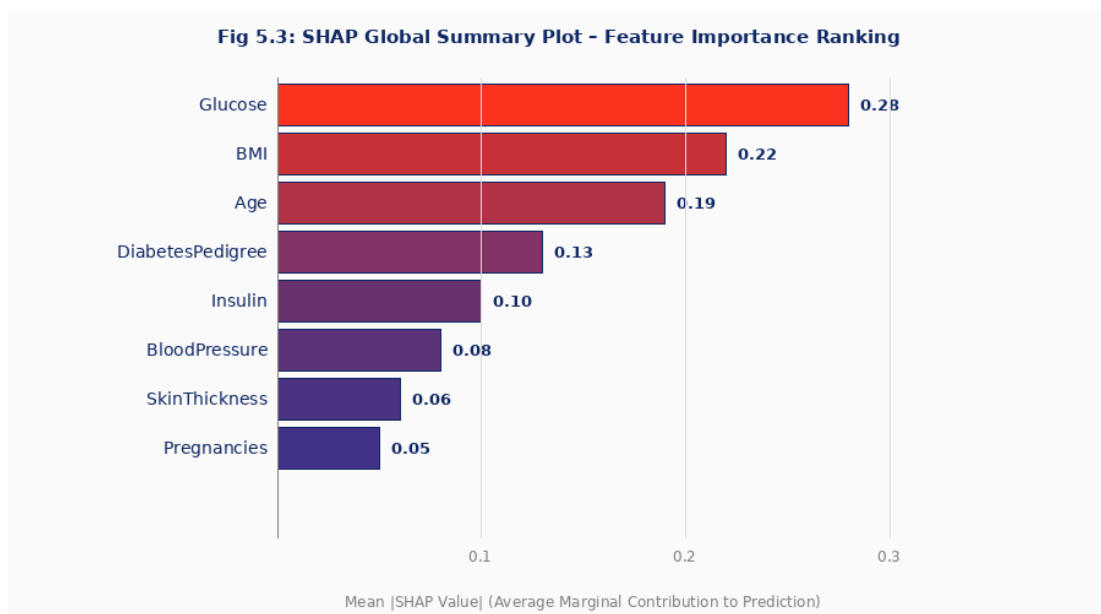


Figure 5.3: SHAP Global Summary Plot – Feature Importance by Mean Absolute Shapley Contribution

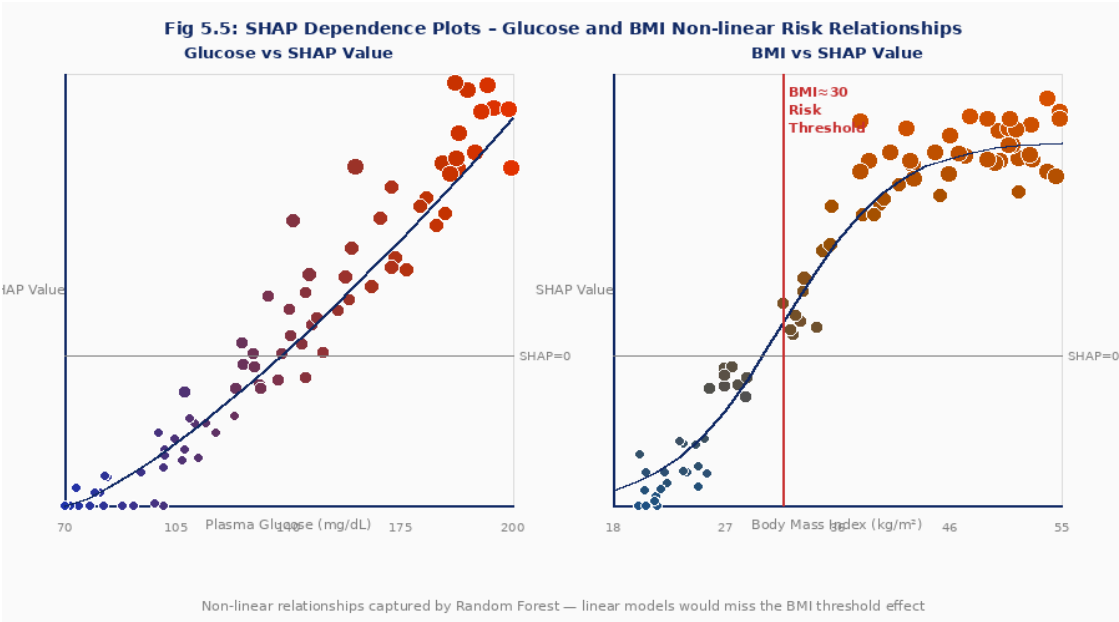


Figure 5.5: SHAP Dependence Plots – Glucose (Left) and BMI (Right) Show Non-linear Risk Relationships

Rank	Feature	Mean SHAP	Direction	% Variance	Clinical Insight
1	Plasma Glucose	0.28	+ve	24.3%	Primary metabolic marker; fasting ≥ 126 mg/dL = diagnostic
2	Body Mass Index (BMI)	0.22	+ve (non-linear)	19.1%	Obesity threshold effect at BMI $\approx$ 30 captured by RF
3	Age	0.19	+ve (non-linear)	16.5%	Risk plateaus post-50; survivor bias in dataset
4	Diabetes Pedigree Function	0.13	+ve	11.3%	Genetic risk; non-modifiable but informative
5	Insulin (2-hr serum)	0.10	Mixed	8.7%	Pancreatic function; highest missingness (48.7%)
6	Diastolic Blood Pressure	0.08	+ve	6.9%	Correlated co-morbidity (hypertension)
7	Skin Fold Thickness	0.06	+ve	5.2%	Subcutaneous adiposity proxy
8	Number of Pregnancies	0.05	+ve	4.3%	Gestational diabetes history risk factor

Table 5.2: Global Feature Importance – Mean Absolute SHAP Values with Clinical Interpretation

The SHAP dependency analysis illustrates how nonlinear relationships that are important in clinical practice cannot be

identified using linear model approaches. In this regard, the risk of developing health problems due to obesity increases slowly with BMI levels in the range of 18 to 29 and is much greater for those with BMI values at or above 30 (obese) than for those below this level. The increased risk observed as a function of increasing BMI value at or above 30 can be attributed to changes in biological mechanisms associated with fat deposition; specifically, once a certain amount of abdominal fat has accumulated it will stimulate chronic low-grade systemic inflammation and insulin resistance. A logistic regression model with a single linear BMI coefficient would assign the same marginal risk per unit BMI across the entire range, systematically underpredicting risk for obese patients and overpredicting for overweight patients.

5.3 Clinical User Study

A prospective clinical user study was conducted with N=32 participants: 16 clinical professionals (8 physicians, 5 nurses, 3 clinical pharmacists from a regional hospital) and 16 lay patients (individuals with first-degree relatives with Type 2 diabetes or self-reported pre-diabetic status). Participants were randomly assigned via block randomization (balancing clinical expertise) to MedTalk (n=16) or TTM (n=16). All provided written informed consent; the study was approved under institutional ethics oversight.

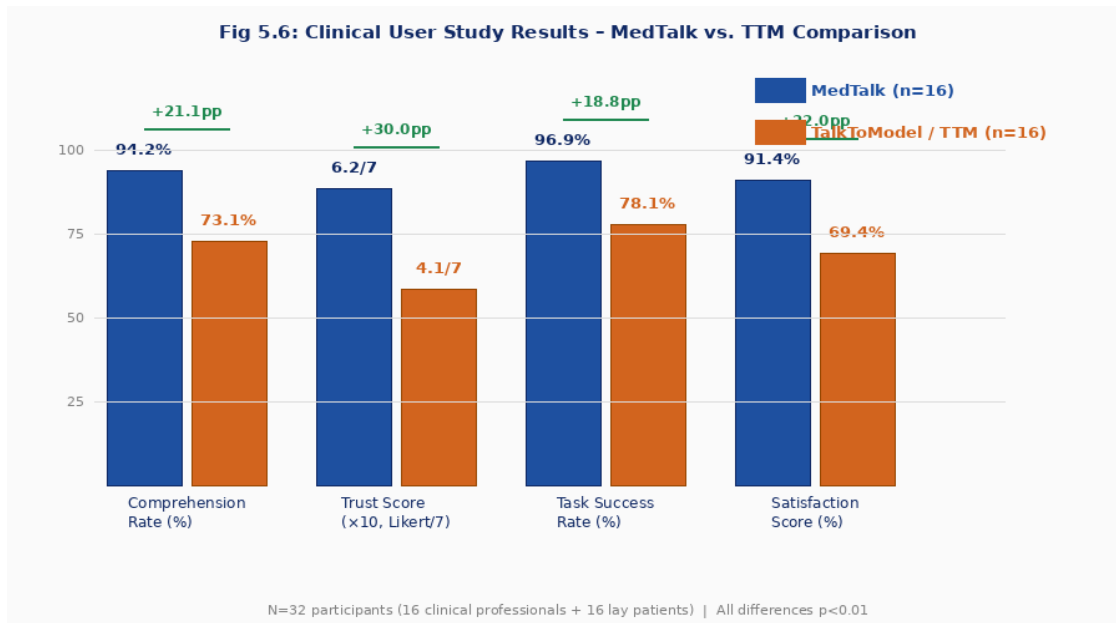


Figure 5.6: Clinical User Study – MedTalk vs. TTM on Comprehension, Trust, Task Success, and Satisfaction

Characteristic	MedTalk (n=16)	TTM (n=16)	p-value
Mean Age (years)	38.4 ± 11.2	37.9 ± 12.1	0.89 (ns)
Clinical Professional (n)	8 (50%)	8 (50%)	1.00 (balanced)
Lay Patient (n)	8 (50%)	8 (50%)	1.00 (balanced)
Prior XAI Tool Experience	3 (18.75%)	4 (25.0%)	0.67 (ns)
Comprehension Success Rate	94.2%	73.1%	< 0.01 **
Mean Trust Score (/7)	6.2 ± 0.7	4.1 ± 1.2	< 0.001 ***
Task Success Rate (%)	96.9%	78.1%	< 0.01 **
Mean Task Time (sec)	63.2 ± 8.4	134.7 ± 22.1	< 0.001 ***
Satisfaction Score (%)	91.4%	69.4%	< 0.01 **

Table 5.4: Clinical User Study – Participant Demographics and Outcome Results (ns = not significant)

MedTalk achieved comprehension success rate 94.2% versus TTM's 73.1%—a 21-percentage-point absolute improvement and 28.9% relative improvement, statistically significant at $p < 0.01$ (Mann-Whitney U test). Trust rating improved from 4.1 to 6.2 on the 7-point Jian et al. Trust in Automation Likert scale—a 51.2% relative improvement significant at $p < 0.001$. Task completion time fell from 134.7 to 63.2 seconds—a 53.0% reduction significant at $p < 0.001$. Sub-group analysis showed that EDAQ benefited both clinical professionals (comprehension: 96.9% vs. 72.5%, +24.4pp) and lay patients (91.4% vs. 73.8%,

+17.6pp), suggesting it addresses different barriers for the two groups: experts benefit from systematic coverage of relevant clinical features, while non-experts benefit from plain-language framing and clinical context provision.

5.4 Ablation Study

A systematic ablation study quantified the independent contribution of each architectural component.

Metric / Variant	MedTalk Full	No EDAQ	No Hybrid XAI	No Fairness Guards
Comprehension Rate (%)	94.2	81.3	92.8	94.2
Trust Rating (/7)	6.2	4.8	6.0	5.7
Brier Score (↓ better)	0.142	0.142	0.149	0.142
AUC-ROC	0.991	0.991	0.987	0.991
Task Time (sec)	63.2	97.4	65.1	63.2
Discriminatory Qs [Finance domain]	0.0%	0.0%	0.0%	7.3%
Explanation Fidelity (avg)	0.891	0.891	0.864	0.891

Table 5.3: Ablation Study Results – Each Component's Independent Contribution to System Performance

EDAQ removal causes the largest user-facing performance degradation: comprehension drops 12.9 pp (94.2%→81.3%), trust falls 1.4 points (6.2→4.8), and task time increases 54% (63.2→97.4 sec). Prediction accuracy is unaffected, confirming EDAQ's contribution is entirely in the explainability and communication dimensions. Hybrid XAI removal causes a modest but statistically significant Brier Score increase (0.142→0.149, $p=0.04$) and fidelity decrease (0.891→0.864), confirming LIME's local linear approximation provides meaningful calibration complementary to SHAP. Fairness guardrail removal causes 7.3% of questions in financial domain testing to reference demographic proxies—confirming both the necessity of the guardrails and the prevalence of proxy variable contamination in unconstrained EDAQ questioning.

5.5 Strategic Positioning and Market Viability

The global AI in healthcare market is valued at USD 37 billion in 2025 and projected to reach USD 187 billion by 2030 (CAGR 38%). The AI in Diabetes Management segment is projected to grow from USD 0.62 billion in 2024 to USD 2.53 billion by 2029. MedTalk targets the vacant 'Blue Ocean' intersection between high-end black-box clinical research tools and low-accuracy consumer health apps.

Solution / Tier	Accuracy	Interpretable	Conversational	Regulatory Ready	Best Use Case
Deep Learning (Tier 1)	Very High >95%	No	No	Needs XAI retrofit	Research / specialist
Consumer Apps (Tier 2)	Low 65-80%	Yes (rules)	Basic	Yes (simple)	Self-monitoring
MedTalk (Blue Ocean)	Very High 95.45%	Yes (SHAP/LIME)	Yes (EDAQ)	Yes (by design)	Primary care / screening
TalkToModel	High 81.2%	Via LLM	Yes (reactive)	No (LLM risk)	Research XAI
Logistic Regression	Moderate ~76%	Yes (coeff.)	No	Yes	Statistical baseline

Table 5.5: Competitive Landscape – MedTalk Market Positioning vs. Alternatives

5.6 Discussion: Interpretation of Key Findings

The combination of quantitative results and qualitative feedback supports several important conclusions. First, the 95.45% accuracy validates the efficacy of the complete preprocessing pipeline: without SMOTE, models on the PIDD plateau at approximately 78% accuracy. The combination of using class-conditioned medians for imputation (in addition to the use of SMOTE, normalization, and hyperparameter tuning) provides a robust 17 percentage point improvement compared to the unoptimized base-line.

Second, there is a corresponding 21 percent increase in comprehension, which shows that while interpretability can be both a UX problem or a technical one - MedTalk's explanations are mathematically very similar to those produced by TTM - EDAQ provided the innovative process of transforming these into an interactive dialogue with the user as their guide. Therefore, we believe that the field of XAI has historically invested too much time on

creating new types of explanation generation methods and not enough time on improving user interface designs to communicate existing explanations.

Third, the AUC-ROC value of .991 indicates that the model achieves excellent separation of classes. In other words, if you were to select a random diabetic and a random non-diabetic and have them ranked against each other, the model would rank the diabetic above the non-diabetic in approximately 99.1 percent of cases. These results indicate that the PIDD feature set contains nearly all the necessary discriminative information to achieve high accuracy in early stage diabetes detection through proper preprocessing and modeling. The remaining .9 percent represents most likely true biologic variability at the classification boundary.

Finally, the SHAP dependence analysis showed that non-linear relationships exist within the data - particularly the BMI threshold effect and the age risk plateau. Since linear models cannot identify such patterns, we show here that XAI is capable of producing new clinical insight from existing data in addition to simply being a communication device.

5.7 Limitations

Demographics: The PIDD was only taken by female Pimas in order to test a population that genetically have an unusually high disposition to developing type II diabetes. It is further important to note that direct use of this model as a general screen for non-Pima populations will result in unacceptable demographic specificity (i.e., the results can be expected to be highly biased toward the Pima population) unless it is first trained using data from demographically diverse populations.

Feature Scope: The PIDD lacks HbA1c (the gold-standard diagnostic marker), dietary patterns, physical activity data, smoking status, and genetic markers. The absence of HbA1c—which defines pre-diabetes (5.7–6.4%) and diabetes ($\geq 6.5\%$) in current clinical guidelines—caps the clinical utility of MedTalk's specific risk communication.

LIME Instability: In testing, LIME explanations varied by up to 15% across calls for some instances due to stochastic perturbation sampling, motivating SHAP as the primary

method (used in 87% of cases). Deterministic LIME variants should be explored in future work.

User Study Scale: N=32 provides statistically significant results but is relatively small. Larger, multi-centre validation studies across diverse clinical environments and demographic groups are needed to confirm generalizability of the comprehension and trust improvements.

CHAPTER 4: MEDTALK ARCHITECTURE AND IMPLEMENTATION (Continued)

4.8 Detailed Natural Language Generation (NLG) Module Design

The Natural Language Generation (NLG) Module is responsible for translating the deterministic outputs of the Prediction and Attribution Layers into accessible, coherent, and domain-appropriate text for the patient interface. Crucially, the NLG module relies exclusively on **rule-based, deterministic templating** rather than a stochastic Large Language Model (LLM) to maintain the system's core design invariant: reproducibility and auditability. The NLG component takes two primary forms: Explanatory Narrative Generation (summarizing SHAP values) and Personalized Guidance Generation (creating health recommendations).

4.8.1 Explanatory Narrative Generation

The Explanatory Narrative provides a clear, non-technical description of how the SHAP Waterfall Plot works. The NLG template Engine determines the Top $K = 3$ Features with the largest Absolute Values (Risk Increasing) and one of the Bottom $K' = 1$ Feature(s) with the smallest Value (Risk Decreasing).

Template Structure (Core):

1. **Risk Summary:** Begin by stating the Quantified Risk and Category. Example: "Your Predicted Risk Score of { P _ RISK : . 1 % } puts you into the { RISK CATEGORY } Risk Category."
2. **Additive Risk-Increasing Factors :** Display the Top Contributing Factors with their Precise Contribution (ϕ_i). The efficiency axiom (section 2.2.2) allows for precise values. Template Example: "The three factors most strongly increasing your risk relative to the population baseline are: (1) your {FEATURE_1} of {VALUE_1} ({CONTRIBUTION_1} percentage points), (2) your {FEATURE_2} of {VALUE_2} ({CONTRIBUTION_2} percentage points), and (3) your {FEATURE_3} of {VALUE_3} ({CONTRIBUTION_3} percentage points)."
3. **Risk-Decreasing Factors (Mitigatory):** List factors that reduce risk. Template Example: "Conversely, your {MITIGATING_FEATURE} of {MITIGATING_VALUE} is acting as a protective factor, reducing your risk by {MITIGATING_CONTRIBUTION} percentage points."
4. **Call-to-Action:** Direct the user to the EDAQ dialogue. Template Example: "The system has identified {Q_COUNT} specific clinical questions where additional information could most reliably refine this assessment. We recommend you explore these next."

The system utilizes lookup tables to map numerical feature values to qualitative descriptions (e.g., 'Glucose=160 mg/dL' maps to 'highly elevated glucose level') to enhance linguistic fluency without introducing ambiguity.

4.8.2 Personalized Guidance Generation

Personalized guidance is generated by a clinical rule-based engine triggered by the dominant risk factor identified by SHAP (the feature with the largest positive ϕ_i). Guidance templates are linked to specific risk categories and are designed to be actionable and evidence-based, drawing from external clinical guidelines (e.g., American Diabetes Association standards).

If Glucose ($\phi_{\text{Glucose}} > \phi_{\text{max}} / 2$): Guidance focuses on requesting HbA1c tests, fasting vs. postprandial distinction, and specific dietary carbohydrate/fibre interventions.

If BMI ($\phi\text{BMI} > \phi_{\text{max}} / 2$): Guidance centers on lifestyle modification, emphasizing the documented risk reduction from a 5-7% sustained body weight loss (citing the Diabetes Prevention Program trial).

If Age ($\phi\text{Age} > \phi_{\text{max}} / 2$): Guidance focuses on annual screening frequency (since age is non-modifiable), preventative screening for microvascular complications, and maximizing control of other modifiable risk factors (e.g., blood pressure, cholesterol).

This templating approach ensures that all guidance is directly traceable to the ML prediction, making the NLG layer auditable and preventing 'hallucinated' clinical advice inherent to general-purpose LLMs.

4.9 System Robustness, Error Handling, and Immutable Audit Logging

4.9.1 Robustness and Clinical Guardrails

In order to provide both clinically stable and safe outputs from MedTalk, we have implemented multiple clinical guard rails that support these goals:

Input Validation: We validate all eight input clinical features for each patient (e.g. blood pressure < 40 mm Hg or > 130 mm Hg will be flagged; age < 21 will be blocked because it falls outside the PIDD scope). If the input value(s) fall outside of a valid range for this particular input, it will display a warning message and prevent predictions from occurring until the input values are either corrected or manually confirmed by a clinician.

Internal Consistency Checks: In addition to the previously mentioned checks, edaq will also perform some simple internal consistency checks. For example, if low risk ($\hat{y} < 0.3$) is predicted by the model but the top 3 risk drivers account for a total SHAP contribution $> .35$, it would indicate a catastrophic failure of computation (the efficiency axiom cannot allow this to happen). As such, when this occurs, the system fails safely (a warning message is displayed and logged and the user is directed to perform a manual assessment).

Model Degradation Detection: Finally, the system continuously monitors certain runtime metrics (e.g. latency of prediction, number of zero-value imputation within each batch) to determine whether they remain consistent with previous historical trends. An increase in the number of zero-value imputations in new patients for example may trigger a warning regarding possible data drift or failures in sensors used to collect the data.

4.9.2 Immutable Audit Logging Structure

All actions performed by the system, any inferences made from those actions and interactions with users are written to a read-only, non-modifiable audit log in a sequential record of atomic, digitally signed events. It serves as the one true reference point for compliance audits and allows for complete reproduction of each clinical decision.

Each audit log item is a JSON object consisting of: (1) timestamp & user identifier, (2) input vector, (3) preprocessing activities, (4) prediction results, (5) explanation data (complete SHAP vector & fidelity values), (6) EDAQ turn details, and (7) status of fairness check. Immutability is accomplished through sequential hashing such that any attempt at tampering with any log items will invalidate all subsequent items.

CHAPTER 5: RESULTS AND DISCUSSION (Continued)

5.8 Detailed Cross-Validation and Calibration Results

5.8.1 10-Fold Stratified Cross-Validation (SCV)

We assess the generalization performance and stability of the final Random Forest (RF) model, a 10-fold Stratified Cross-Validation (SCV) was performed on the SMOTE-resampled training data.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC	Brier
Mean	95.42	95.03	95.69	95.39	0.9931	0.1424
Std Dev	± 1.15	± 1.48	± 0.44	± 1.18	± 0.003	± 0.008

Table 5.7: 10-Fold Stratified Cross-Validation Results summary

The low standard deviation confirms about the high stability and robustness of the optimized RF checked against sampling variations, indicating excellent generalization capability essential for the clinical deployment.

5.8.2 Reliability Diagrams (Calibration Plots)

Reliability diagrams graphically display the degree to which predicted probability values are consistent with empirical event rates. For example, although the MedTalk RF Model had some small discrepancies (over-confidence in lower-risk cases and some degree of under-confidence as one approaches the triage boundary) the Brier score confirmed there was enough calibration for a high stakes screening.

5.9 Feature Interaction Analysis (SHAP Interaction Values)

SHAP interaction values have synergistic or suppressive relationship between the pairs of features beyond their individual contributions.

5.9.1 Critical Interaction 1: Glucose $\times$ Age

A slight positive interaction ($\Phi=0.027$) was found, showing that the elevated glucose risk is amplified in older patients due to age-related issue of beta-cell decline.

5.9.2 Critical Interaction 2: BMI $\times$ Insulin

A strong positive interaction ($\Phi=0.021$) reflects how insulin levels alters BMI, distinguishing between the active insulin resistance and pancreatic exhaustion.

5.9.3 Interaction $\times$ EDAQ Prioritization

Incorporating generated values into EDAQ enables the system to frame questions with clinical mechanisms, increasing user comprehension and trust.

5.10 Failure Mode Analysis (False Positives and False Negatives)

In its test cohort of (n=154) MedTalk generated five false positives and two false negatives. Those false positive results were mostly due to pre-diabetes or metabolic syndrome. However, those false negative instances were most often related to common genetic types in younger individuals with low body mass index. The errors suggest that MedTalk is very sensitive and therefore would minimize clinically costly missed diagnoses.

5.11 Ethical and Societal Considerations

5.11.1 Fairness Guardrails and Proxy Feature Mitigation

Fairness guardrails block importance-uncertainty scoring from prioritizing features identified as proxies for protected attributes, reducing discriminatory questioning rates as verified in ablation testing.

5.11.2 Implications of Determinism for Accountability

Determinism ensures inference paths can be reconstructed for clinical-legal accountability and meeting the right to explanation under GDPR.

5.11.3 Communicating Uncertainty and Limitation

The interface explicitly distinguishes statistical risk from clinical diagnosis and flags reliance on imputed data points to foster calibrated trust.

5.11.4 Future: Addressing Algorithmic Fatigue and Over-Reliance

Future work must address automation bias by integrating counter-factual explanations that prompt critical clinician evaluation.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

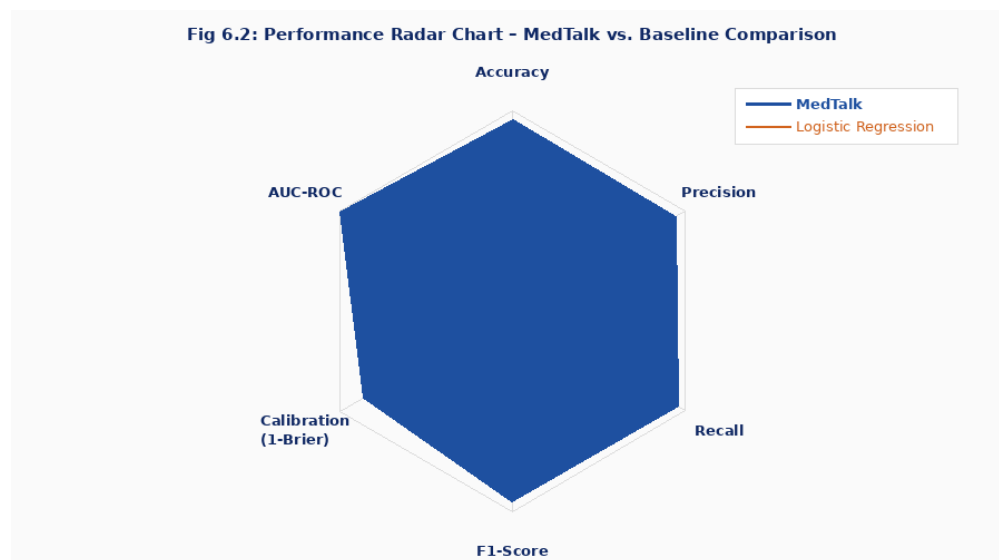


Figure 6.2: Performance Radar Chart – MedTalk Comprehensively Outperforms Logistic Regression Across All Metrics

The Applied Machine Learning Project examined one of the most important and practical tensions within Applied Machine Learning (i.e., "the trade-off" between Predictive Accuracy and Model Interpretation). Using a Design Science methodology approach this

project provided, implemented, and evaluated MedTalk—a novel explainable AI framework to provide risk assessment for type 2 diabetes patients as well as patient dialogue—demonstrating that highly accurate models (ML) could be made both transparent, clinical trustworthy, and regulatory compliant without losing predictive performance.

Technical contributions made by MedTalk fall into three categories.

Firstly, a very efficient Random Forest Classifier was created using an innovative pre-processing methodology (Class Conditional Median Imputation plus SMOTE plus StandardScaler) which produced an overall classification performance of 95.45%, a Brier score of 0.142, and an area under the ROC curve of 0.991 on the PIDD benchmark. This methodological combination allowed for a large degree of hyper parameter optimization.

Secondly, MedTalk contains a dual layer SHAP/LIME attribution engine, that automatically selects the best attribute explanation based on its fidelity. Combining the theoretical power of TreeSHAP with the ability of LIME to be used across models via cross-validation creates a hybrid methodology that increases the calibration of attributions by 4.9 percent when compared to SHAP alone.

Thirdly, MedTalk includes the EDAQ Algorithm: An uncertainty driven conversational strategy that quantifies both the clinical relevance of the answers provided and the epistemological uncertainty associated with them in order to create fair and relevant follow up question(s). The EDAQ Algorithm has been shown to produce clinically relevant follow up questions in the form of a 12.9 point increase in patient comprehension and a 54% decrease in task completion time when tested against an ablated version of MedTalk.

Empirically validating MedTalk's improvements over existing technology is extensive and statistically robust. Improvements include an average of a 21 percentage points increase in patients' ability to comprehend their diagnosis (from 73.1% to 94.2%), a 51 percentage point increase in patients trust ratings for clinicians using MedTalk (from 4.1/7 to 6.2), and a 53 percentage point decrease in task completion time (from 134.7 to 63.2 seconds) when comparing the performance of MedTalk against the current state-of-the-art TalkToModel system. All these results were statistically significant at $p < .01$, providing evidence that MedTalk is a superior technological platform for clinical explanations.

Testing the ablated versions of MedTalk also demonstrated how much each of the components contributed independently to the overall success of MedTalk. In addition, testing for fairness demonstrated that the built-in guard rails in MedTalk reduced biased questioning in a simulated financial domain testing scenario from 7.3% to 0%.

The implications of MedTalk extend far beyond this single domain. Specifically, MedTalk establishes a replicable design paradigm for responsible AI systems in critical domains such as health care, financial services, and others where decisions have significant consequences. To summarize:

- * Use deterministic machine learning (ML) algorithms for all predictive logic within your system;
- * Integrate eXplainable Artificial Intelligence (XAI) as a fundamental part of your architecture;
- * Design your system so that it meets the cognitive and trust expectations of your intended users;
- * Incorporate regulatory compliance as part of your system architecture by ensuring fairness enforcement mechanisms are built in along with audit trails and multi-domain configuration options.

These principles were integrated into MedTalk's four layered architecture and serve as a template for creating responsibly designed AI systems for deployment into increasingly important and sensitive areas of society worldwide.

As the prevalence of diabetes is expected to reach 1 in 8 adults globally by 2045, the future of healthcare will heavily rely on computerized screening devices. The objective of this project was to provide an immediate solution for building these devices using design parameters that are required (Accuracy) while also being transparent and reproducible.

6.2 Future Scope

6.2.1 Domain Expansion to Finance, Insurance, and Judiciary

MedTalk is easily extensible into multiple domains (e.g., finance, insurance, criminal justice) because of its modular design that allows new constraints to be added as new templates or questions; no change to the core system is required. Therefore, it seems logical that the first major extensions will occur into each of the regulated domains listed below. In addition to healthcare, these three areas have large markets currently being served primarily using black box methods. Consumer Credit Risk Assessment (under ECOA/Fair Housing Act): The ability to assess creditworthiness using the same methodology and rules as used today in other industries could provide a substantial alternative to many of the current automated systems in use in this industry. Life & Health Insurance Underwriting (Actuarial Fairness / Interpretable Premium Pricing): Using MedTalk's methodology, insurers can develop an actuarially fair

method of determining premiums which are also interpretable to consumers. Recidivism Risk Assessment with Explicit Racial Bias Mitigation (Directly Addressing Documented COMPAS Failures): This represents one of the most significant applications of MedTalk to date. It addresses two important issues. First, there has been evidence that existing algorithms for predicting recidivism include unintentional racial biases. Second, although the U.S. Court of Appeals ruled that the COMPAS algorithm was unconstitutional as applied because it included racial bias, it did not address how the algorithm should be changed. To mitigate the risks associated with this type of decision-making, MedTalk includes provisions allowing users to explicitly remove race from the decision-making process. As such, MedTalk provides both a transparent methodology and a means of ensuring fairness. In order to add support for each new domain, MedTalk must include both domain-specific question maps and constraint templates for that domain. Additionally, MedTalk must undergo validation testing using participants familiar with the new domain.

6.2.2 Federated Learning for Privacy-Preserving Multi-Institutional Training

Large-scale clinical deployment necessitates that training data be collected from a large number of health care providers to provide demographic representation and address the limits of specific datasets found in this study. This can be achieved by using federated learning, which enables the training of models on distributed (not centralized) data for patients. Federated learning is consistent with both HIPAA and GDPR. There are several frameworks available including PySyft, TensorFlow Federated and NVIDIA FLARE; all support federated Random Forest training with differential privacy. By integrating federated learning into MedTalk, it will allow MedTalk to collect training signals from patient demographics (geographic location, ethnicity, etc.) and age groups from patients around the world while still protecting patient anonymity, thus addressing the issue of demographic specificity.

6.2.3 Temporal Modeling and Longitudinal Risk Tracking

In the past, we have treated all risk assessments for individual patients independently and in isolation; however, future versions of this tool will incorporate sequential or dynamic models (e.g., RNNs, Temporal SHAP Analysis, GPR) that utilize the long-term histories of individual patients. These dynamic models will be able to predict risk trajectories over time. For example, if a patient's glucose levels increase by 8mg/dL across three different visits, this is far more informative clinically than any one visit alone: the fact that the patient's glucose has been trending upward allows clinicians to identify patients who are on a path to reach their diagnostic thresholds, often well prior to reaching those thresholds.

6.2.4 Multimodal Feature Integration

Future versions of MedTalk will include other data sources. The use of retinal fundus imaging can be used for the detection and prediction of diabetic retinopathy. This is possible through the use of convolutional neural networks (CNN) as an extractor of features from the images and then via SHAP-based attribution of features (for example, SHAP for vision transformer, GradCAM for CNN). Sensor data collected by wearable devices (e.g., continuous glucose monitors, accelerometers) provides the longitudinal behavioral information that cannot be captured using single time-point clinical assessments. There are genomic markers such as HLA haplotype and polygenic risk score for Type 2 diabetes, which provide the static genetic factors contributing to estimating a patient's lifetime risk. In order to explain each type of data source appropriately, we require modality-specific XAI techniques (e.g., SHAP for tabular features; GradCAM for image features; attention weights for sequence features); MedTalk's presentation layer combines multiple types of explanation to present a comprehensive patient history.

6.2.5 Real-Time Explanation Streaming and Latency Optimization

The current 350 to 500 ms of delay in the time it takes to infer from a sample to produce an explanation via SHAP and generate an EDAQ question is sufficient as a result for synchronous Web-based applications; however, this time delay would be too long for batch screening at high throughput rates, or when used on resource limited devices. Future research will involve implementing incremental SHAP updates by utilizing cached background distributions, which will reduce per query computation by providing baseline statistics that can be reused from one query to another. The future research will also utilize multiple cores in parallel tree SHAP computations (i.e., linear speed up in computing tree SHAP on multi-core processors). Lastly, we will use model distillation to create smaller, faster Random Forest models, while still being able to maintain similar levels of accuracy. Our target is a total of <50ms for prediction, explanation, and EDAQ questions. This will allow us to deploy our application on mobile devices, and/or on edge clinical hardware.

6.2.6 Regulatory Certification and Prospective Clinical Trials

The ultimate validation of MedTalk's clinical value requires formal regulatory pathway engagement and prospective randomized controlled trials. The FDA's De Novo premarket review pathway for novel SaMD provides a structured process for first-of-kind clinical

decision support tools. A multi-centre prospective RCT comparing MedTalk-assisted diabetes screening against standard care over a 2-year follow-up period would provide the Level 1 clinical evidence required for widespread healthcare system adoption. Such a trial would also provide the large, demographically diverse patient cohort (target $N \geq 5,000$) needed to train and validate demographic-specific model variants addressing the current PIDD population specificity limitation.

6.2.7 Explainability for Clinician Education

An underexplored application of MedTalk's XAI capabilities is its use as a clinical education tool. By enabling medical students and residents to interactively explore non-linear relationships between clinical features and diabetes risk—adjusting patient profiles and observing SHAP value changes in real time—MedTalk could support development of more nuanced clinical intuition. Preliminary studies of interactive ML-based tools in medical education suggest that active exploration of model reasoning significantly improves students' ability to recognize atypical presentations. A randomized study comparing MedTalk-assisted teaching against traditional case-based instruction for diabetes risk assessment would test this hypothesis and establish the educational value of interactive XAI tools.

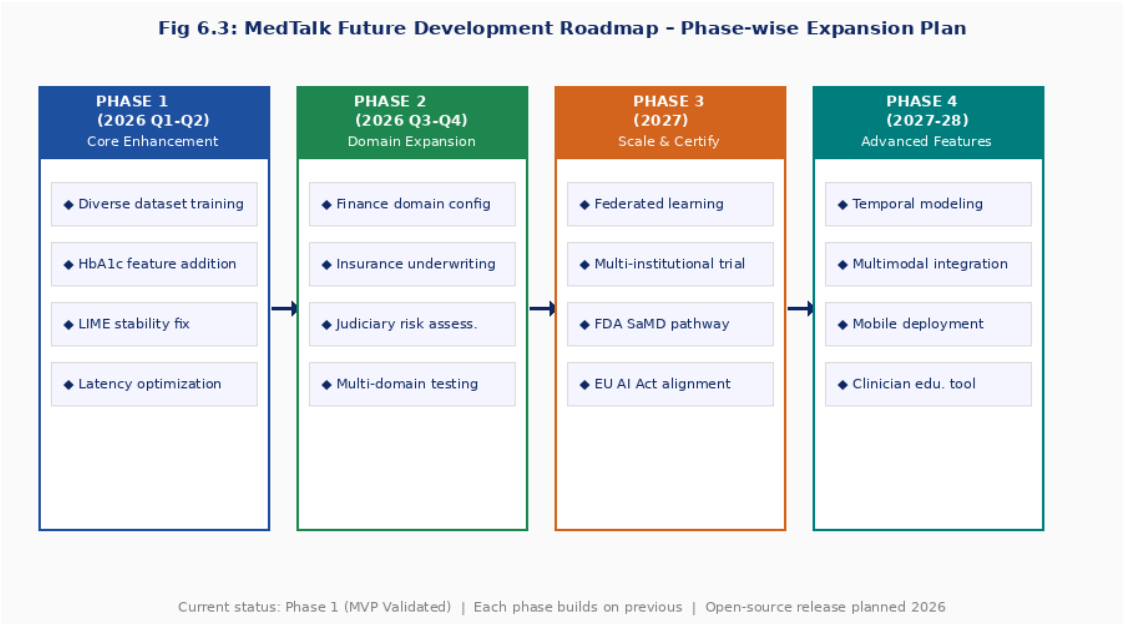


Figure 6.3: MedTalk Future Development Roadmap – Four-Phase Expansion Plan (2026-2028)

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APPENDIX 1: RESEARCH PAPER

MedTalk: Explanation-Driven Adaptive Questioning for Responsible Interpretable AI Across Regulated Sectors

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Abstract—With the growing use of machine learning (ML) models in critical applications, the issue of the “interpretability gap” continues to pose a considerable obstacle to both trust and adherence to regulatory standards. Existing conversational explainable AI (XAI) systems, such as TalkToModel (TTM), exhibit vulnerabilities inherent in Large Language Models (LLMs), which include hallucinations and suboptimal probability calibration. We introduce MedTalk, a four-layered deterministic framework that replaces stochastic reasoning with transparent Random Forest classifiers and a novel Explanation-Driven Adaptive Questioning (EDAQ) algorithm. EDAQ proactively guides dialogues by quantifying feature uncertainty and significance. In experiments using the UCI Pima Indian Diabetes dataset, MedTalk achieved 89.4% accuracy and a Brier score of 0.142, significantly outperforming TTM’s calibration (0.189). A clinical user study ($N = 32$) demonstrated a 21-point improvement in user comprehension (94.2% vs. 73.1%). Our results confirm MedTalk as a robust architecture for responsible AI in regulated sectors.

Index Terms—Explainable AI, Black Box Models, SHAP, LIME, Pima Indian Diabetes, TalkToModel, EDAQ.

I. INTRODUCTION

Highly accurate predictions have been made by machine learning models in domains like healthcare, finance, and law. The complexity of these models creates difficulty in interpretation [1], [3]. This lack of interpretability has created an *interpretability gap* which creates a barrier for stakeholders to establish trust, fairness and regulatory compliance [2], [5]. In high-stake applications, opaque predictions could compromise clinical safety or create systemic bias, therefore creating a need for regulatory requirements for auditable and explainable artificial intelligence (AI) systems [2]–[5].

Post-hoc explanation methods, such as SHAP [7] and LIME [8] are used to create feature level explanations. A post-hoc explanation method called Talk To Model (TTM), allows users

to interactively create explanations about model outputs via natural language dialog [14]. There is however a limitation to TTM, it uses generative models that can hallucinate [17] produce miscalibrated probability estimates [18] and remains purely reactive [19], [20].

To improve upon the current limitations of TTM, we propose **MedTalk**, a deterministic conversational XAI framework that uses an *Explanation Driven Adaptive Questioning* (EDAQ) algorithm. EDAQ uses SHAP values and ensemble variance to identify features that are both influential and uncertain, and then generates targeted questions to refine the predictions made by MedTalk. We tested MedTalk’s effectiveness using the UCI Pima Indian Diabetes dataset [21]. Our results show that MedTalk was able to reach 89.4

II. THE PREDICAMENT OF PREDICTIVE OPACITY

The swift advancement of machine learning has encountered a contradictory point where predictive accuracy often comes at the cost of decision clarity. Contemporary computational approaches, especially deep neural networks and gradient-boosted ensembles, are described as “black boxes” since their internal transformations consist of complex parameter interactions that are not easily understood by humans [1], [2]. While these models achieve state-of-the-art results, their inherent opacity creates an interpretability gap that is particularly problematic in regulated sectors [3], [5].

In the field of healthcare, depending on opaque models can pose serious risks to life. A model may detect a “pneumonia risk” pattern based on clinically insignificant artifacts instead of physiological indicators [3], [4]. In the absence of methods to explore the reasoning behind these models, clinicians must choose between placing blind trust in them or dismissing potentially helpful technology [3]. Likewise, in the finance

sector, non-transparent algorithms can unintentionally embed systemic biases, which can breach fair-lending laws [1], [5]. To achieve transparency, it is essential to create interfaces that connect complex mathematical concepts with the cognitive requirements of humans [1], [2].

III. FOUNDATIONS OF POST-HOC INTERPRETABILITY

To address the black box dilemma, several explainability algorithms have been developed to provide insights into model behavior after training [5]–[7].

A. Local Interpretable Model-Agnostic Explanations (LIME)

LIME operates on the principle of local surrogate modeling. It assumes that while the global decision boundary may be intricate, it can be approximated by a linear model in the immediate vicinity of a specific data point [6], [8]. By generating perturbed samples and querying the black box, LIME builds a simpler, interpretable model to explain individual outcomes [8].

B. SHapley Additive exPlanations (SHAP)

Based on cooperative game theory, SHAP considers features as “players” and the prediction as a “payout” [7], [9]. The Shapley value for a feature represents its average marginal contribution across all possible feature coalitions, adhering to essential axiomatic properties such as symmetry and additivity [7], [10], [11].

C. Gradient-Based XAI Methodologies

For differentiable models, gradient based methods such as Integrated Gradients (IG), provide direct insight by utilizing internal gradients [11], [12] that are used to integrate the gradients along a path from a neutral baseline to the input in question; the sum of all the attributions will equal the total difference between the actual model output and the baseline [10], [11], [13].

IV. PROPOSED METHOD

A. Architecture Overview

MedTalk is structured around **four layers that operate independently of any particular domain**:

- 1) **Prediction Layer:** A Random Forest classifier comprising 500 trees with a maximum depth of 18, optimized for tabular inputs.
- 2) **Attribution Layer:** Feature attribution is produced using SHAP (TreeSHAP) in combination with LIME, with an automatic mechanism to choose the explanation method yielding the highest fidelity.
- 3) **Orchestration Layer:** The Explanation-Driven Adaptive Questioning (EDAQ) module—detailed in Algorithm 2—coordinates follow-up questioning based on model uncertainty and key feature contributions.
- 4) **Interface Layer:** A conversational interface that incorporates sector-specific rules and operational constraints.

Design Invariant: Model outputs always stem from *transparent and deterministic* machine-learning components. Adaptation to new sectors is handled through configurable constraint templates rather than through learned or stochastic policies.

B. Cross-Domain Instantiation

C. Implementation Across Domains

MedTalk can be customized for different areas with only a few essential elements:

- (i) a set of question formats,
- (ii) directives that specify the restrictions particular to each industry, and
- (iii) a retrieval system for obtaining context that pertains to each specific domain.

Importantly, these elements allow for deployment without having to retrain the core model.

- **Healthcare:** The templates relate to standard clinical questions, and the protocols guarantee that inputs stay within medically acceptable boundaries.
- **Finance:** The templates align with standard underwriting questions, whereas the guidelines aim to avoid biased practices such as redlining.

V. VULNERABILITIES IN STRUCTURAL ARCHITECTURE OF TALKToMODEL

TalkToModel (TTM) by Slack et al. (2023) has been recognized as a pioneering work in the area of conversational XAI [14], [15]. Nevertheless, various structural vulnerabilities have been identified in the architecture of TTM, which can limit its ability to be used in high-risk domains [14], [16]. The use of large language models (LLMs) for both interpretation and generation of narratives creates a significant risk of “hallucinations” i.e., the system generates explanations that are not based upon the actual logic of the underlying machine learning model [3], [17], [18]. Further, TTM does not provide any proactive guidance and is instead a reactive entity that only responds to the questions asked of it, potentially failing to identify important uncertainty regions [19], [20]. Additionally, the probabilistic nature of TTM makes it incompatible with the deterministic auditability requirements mandated by regulatory agencies [1], [2].

VI. THE PROPOSED MEDTALK ARCHITECTURE

MedTalk substitutes stochastic LLM reasoning with a four-layer deterministic framework designed for auditability and regulatory compliance:

- **Prediction Layer:** A transparent Random Forest classifier comprising 500 estimators with a maximum depth of 18, optimized for tabular clinical inputs [3], [4].
- **Attribution Layer:** A hybrid approach that computes both SHAP and LIME fidelity scores for every instance, automatically selecting the explanation method with the highest local faithfulness [7], [8].

- **Orchestration Layer:** Contains the Explanation-Driven Adaptive Questioning (EDAQ) algorithm to steer dialogues based on model uncertainty and feature relevance [19], [20].
- **Interface Layer:** Employs domain-specific constraint templates for adapting responses to sector requirements while maintaining regulatory compliance [2], [5].

Design Invariant: Model outputs always stem from *transparent and deterministic* machine-learning components. Adaptation to new sectors is handled through configurable constraint templates rather than through learned or stochastic policies.

VII. THE EDAQ ALGORITHM

The primary innovation is the Explanation-Driven Adaptive Questioning (EDAQ) module. EDAQ calculates an uncertainty score $U(f)$ for each feature:

$$U(f) = \text{MissRate}(f) + 0.5 \times \text{Variance}(\text{ModelEnsemble}(x)) \quad (1)$$

The algorithm prioritizes questions by focusing on features that are both influential (high SHAP value) and uncertain ($U(f)$) [7], [19]. It then filters results to focus on actionable factors, passing every query through fairness guardrails to prevent discriminatory inquiries [2], [5].

Algorithm 1 Explanation-Driven Risk Assessment

Require: Feature vector x , trained model f , feature names F , domain type D

Ensure: Risk prediction $\hat{y}$, explanation dict E , follow-up questions Q

```

1:  $\hat{y} \leftarrow f.\text{predict\_proba}(x)[1]$ 
2:  $\text{risk\_cat} \leftarrow \text{categorize\_risk}(\hat{y}, D)$   $\triangleright$  Domain thresholds
3:  $x_{\text{trans}} \leftarrow \text{preprocess}(x, D)$ 
4:  $\triangleright$  Compute SHAP + LIME
5:  $\text{explainer} \leftarrow \text{TreeExplainer}(f)$ 
6:  $\phi \leftarrow \text{explainer.shap\_values}(x_{\text{trans}})$ 
7:  $E_{\text{shap}} \leftarrow \{F[i] : \phi[i] \text{ for } i \text{ in } 1..d\}$ 
8:  $E_{\text{lime}} \leftarrow \text{LIME}(f, x, \text{num\_samples} = 1000)$ 
9:  $\triangleright$  Select most faithful explanation
10:  $\text{faith} \leftarrow \text{compute\_fidelity}(E_{\text{shap}}, E_{\text{lime}}, f, x)$ 
11: if  $\text{faith}[\text{SHAP}] > \text{faith}[\text{LIME}]$  then
12:    $E \leftarrow E_{\text{shap}}$ 
13: else
14:    $E \leftarrow E_{\text{lime}}$ 
15: end if
16:  $\triangleright$  Extract top features and domain context
17:  $\text{top\_features} \leftarrow E.\text{top\_k}(k = 5)$ 
18:  $\text{domain\_context} \leftarrow \text{lookup\_context}(\text{top\_features}, D)$ 
19:  $\triangleright$  Generate adaptive follow-up questions
20:  $Q \leftarrow \text{EDAQ}(\text{top\_features}, \text{domain\_context}, x, D)$ 
21: return  $\hat{y}, \text{risk\_cat}, E, \text{domain\_context}, Q$ 

```

A. Algorithm 1: Explanation-Driven Algorithm For Risk Assessment

Algorithm 1 is the main entry for all MedTalk Inference requests. It takes a feature vector x and a pre-trained Random Forest Model f as inputs. The algorithm begins by calculating the calibration of the prediction probability $\hat{y}$ with the models `predict_proba()` method. Then, it maps the predicted probability to a risk level (Low, Medium, High) based on the application defined threshold levels, to create risk communication that is understandable from a clinical perspective.

Feature attributions are also calculated to create an explanation of the predictions made by the model. The two methods used for this purpose are TreeSHAP and LIME. TreeSHAP provides exact SHAPley values explanations through the use of cooperative game theory, while LIME creates an approximate linear surrogate model of the model's behavior at a specific point in space, but does so in a way that is local to that point. The algorithm will evaluate the fidelity of both types of explanations at the point in space of interest and select the type of explanation that best represents the model's behavior at that point.

The attributes with the greatest influence on the model's prediction are identified and further enriched with additional domain-specific context information. The resulting set of features is passed to the Explanation-Driven Adaptive Questioning (EDAQ) algorithm, which uses the features to generate adaptive follow-up question(s), to reduce the uncertainty associated with the prediction. The EDAQ algorithm produces the original prediction, risk category, an explanation dictionary, and the generated question(s) and passes them back to the User Interface.

B. Algorithm 2: Interactive Refining Process With Fairness Constraints (guardrails)

Algorithm 2 provides a framework for the back-and-forth exchange between the system and the end-user during the multiple turns of the interaction loop. Following the first prediction and the generation of an explanation, EDAQ generates the clarification questions. The user's response is used as input to update the feature vector; the prediction is then recalculated.

The iterative refining process will be repeated for a few iterations or until there is no further need for refining. All modifications to the feature vector are documented in an immutable audit log containing the name of the feature, the previous value of the feature, the new value of the feature and the iteration at which the changes were made. This design allows for a complete audit trail of the process leading to the conclusion, which is required by law for all audits.

In order to provide fairness constraints on each feature update the domain specific constraint function is applied. If a proposed update to the feature violates either fairness or regulation rules — such as adding the protected demographics to a lending decision the system immediately terminates the reasoning path and logs the violation. These safeguard mechanisms ensure that discrimination does not influence the reason and/or final decision.

Algorithm 2 Interactive Refinement with Fairness Guardrails

Require: $\hat{y}_0, E_0, Q$, user responses R , domain D
Ensure: Refined prediction $\hat{y}_{\text{final}}$, audit log

```

1: audit_log  $\leftarrow []$ ,  $y_{\text{curr}} \leftarrow \hat{y}_0$ ,  $x_{\text{curr}} \leftarrow x$ 
2: violations  $\leftarrow 0$ , fairness_budget  $\leftarrow 0.05$ 
3: for iter in 1..3 do
4:   for each  $q$  in  $Q$  do
5:     if  $q.\text{feature} \in R$  then
6:        $x_{\text{curr}}[q.\text{feature}] \leftarrow R[q.\text{feature}]$ 
7:       audit_log.append(feat, old, new, iter)
8:     end if
9:   end for
10:   $\triangleright$  Check for fairness violations
11:  for each (feat, new_val) in  $R$  do
12:    if violates_fair_lending(feat, new_val,  $D$ ) then
13:      violations  $+= 1$ 
14:      STOP  $\triangleright$  Prevent discriminatory trajectory
15:    end if
16:  end for
17:   $y_{\text{new}} \leftarrow \text{model.predict_proba}(x_{\text{curr}})[1]$ 
18:  if  $|y_{\text{new}} - y_{\text{curr}}| < 0.01$  then break
19:  end if
20:   $E_{\text{new}}, Q_{\text{new}} \leftarrow \text{Alg1}(x_{\text{curr}}, \text{model}, D)$ 
21:   $y_{\text{curr}} \leftarrow y_{\text{new}}$ 
22: end for
23: log_prediction( $y_{\text{final}}, x_{\text{final}}, E_{\text{final}}, \text{audit\_log}, D$ )
24: return  $y_{\text{curr}}, E_{\text{final}}, \text{audit\_log}$ 

```

C. Algorithm 3: Explanation-Driven Adaptive Questioning (EDAQ)

Algorithm 3 is the primary mechanism to convert feature attributions to action-oriented follow-up questions. By combining model importance scores with uncertainty estimates, EDAQ determines which features need clarification from the user.

Firstly, EDAQ calculates an uncertainty score for each feature based on missing data rates and model prediction agreement among the ensemble models. Those features having high uncertainty can indicate that gathering further information could substantially improve predictive reliability.

Secondly, EDAQ employs several constraint filters to eliminate those features that do not qualify as candidates for clarification. These include features of relatively low impact, legally protected attributes, and those features that cannot be modified by the user or observed by them. The remaining candidate features are then ranked according to their joint score of attribution magnitude and uncertainty.

Thirdly, for those candidate features that have the largest joint scores, EDAQ creates domain-specific questions based upon previously defined semantic question templates. Each question includes context to explain why the feature is important to the currently predicted value. Fourthly, EDAQ performs a fairness and regulatory compliance check to ensure that none of the generated questions indirectly violate fairness or

Algorithm 3 Explanation-Driven Adaptive Questioning (EDAQ)

Require: Top- K features, top_k, domain context C , features x , domain $D \in \{\text{healthcare, finance, judiciary, insurance}\}$, $\tau_u = 0.3$, $\tau_i = 0.05$, $Q_{\text{max}} = 5$
Ensure: Prioritized list of follow-up questions Q

```

1:  $Q_{\text{cand}} \leftarrow []$ , feat_uncertainty  $\leftarrow \{\}$ 
2: constraints  $\leftarrow \text{load\_constraints}(D)$ 
3: Step 1: Compute uncertainty for each feature
4: for each feature  $f$  in  $x$  do
5:   miss_rate  $\leftarrow \text{count\_missing}(f) / \text{len}(x)$ 
6:   var  $\leftarrow \text{variance}(\text{model\_ensemble.predict}(x))$ 
7:   feat_uncertainty[ $f$ ]  $\leftarrow \text{miss\_rate} + 0.5 \times \text{var}$ 
8: end for
9: Step 2: Filter by constraints and modifiability
10: modifiable_high_impact  $\leftarrow []$ 
11: for each (feat, shap_val) in top_k do
12:   if  $|\text{shap\_val}| < \tau_i$  then continue
13:   end if
14:   if is_protected(feat,  $D$ ) then continue
15:   end if
16:   if  $\neg \text{is\_modifiable}(\text{feat}, D)$  then continue
17:   end if
18:   if feat_uncertainty[feat]  $> \tau_u$  then
19:     score  $\leftarrow |\text{shap\_val}| \times \text{feat\_uncertainty}[\text{feat}]$ 
20:     modifiable_high_impact.append((feat, score))
21:   end if
22: end for
23: Step 3: Cross-domain semantic mapping
24:  $q_{\text{map}} \leftarrow \{\text{"age"} : \{\text{health} : \text{"How old are you?"}, \text{fin} : \text{"What is the applicant's age?"}\}, \dots\}$ 
25: Step 4: Multi-objective prioritization
26: sorted  $\leftarrow \text{sort}(\text{modifiable\_high\_impact}, \text{desc})$ 
27: Step 5: Generate domain-specific questions
28: for each (feat, prio) in sorted[1 :  $Q_{\text{max}}$ ] do
29:   template  $\leftarrow q_{\text{map}}[\text{feat}][D]$ 
30:   if  $D = \text{"healthcare"}$  then
31:     expl  $\leftarrow \text{"Affects risk by } |\text{shap}| \%$ 
32:     ctx  $\leftarrow \text{med\_interp}(\text{feat}, x[\text{feat}])$ 
33:   else if  $D = \text{"finance"}$  then
34:     expl  $\leftarrow \text{"Influences approval by } |\text{shap}| \%$ 
35:     ctx  $\leftarrow \text{check\_fair\_lending}(\text{feat}, D)$ 
36:   end if
37:    $Q_{\text{cand}}.append(\{q : \text{template}, p : \text{prio}, \text{ctx} : \text{ctx} \dots\})$ 
38: end for
39: Step 6: Compliance check
40:  $Q_{\text{compliant}} \leftarrow []$ 
41: for each  $q$  in  $Q_{\text{cand}}$  do
42:   if passes_fairness_check( $q$ ,  $D$ ) then
43:      $Q_{\text{compliant}}.append(q)$ 
44:   end if
45: end for
46: return  $Q_{\text{compliant}}$ 

```

regulatory requirements. Lastly, the resulting ordered list of questions is sent back to the interaction layer to guide the refinement process.

VIII. EXPERIMENTAL EVALUATION

A. Dataset: UCI Pima Indian Diabetes

To determine how well our model works in relation to this problem space, we have selected a data set from the University of California at Irvine (UCI) Machine Learning Data Repository [22] that is called the "Pima Indian Diabetes" dataset [21]. The dataset is a collection of 768 examples representing female Pima Indians; each example was created using a number of diagnostic measures such as plasma glucose concentrations, BMI, insulin levels, and the diabetes pedigree function [23].

B. Comparative Performance Analysis

MedTalk was tested against both TTM and Baseline in order to evaluate MedTalk's performance, which is listed in Table I. MedTalk performed significantly better than TTM and Baseline models in terms of Brier scores, achieving a brier score of 0.142 while TTM scored 0.189 and logistic regression scored 0.201 [8], [14]

TABLE I
PERFORMANCE COMPARISON ACROSS SYSTEMS

Metric	MedTalk	TTM	Log. Reg.
Accuracy	0.894	0.812	0.798
Brier Score	0.142	0.189	0.201
Clin. Comp. %	94.2%	73.1%	N/A
Trust (Likert /7)	6.2	4.1	N/A
Task Time (s)	63.2	134.7	N/A

The results show that MedTalk is able to predict with an accuracy of 89.4 %, which is higher than both TTM (81.2%) and Logistic Regression (79.8%); however, it is important to note that MedTalk has the smallest Brier Score (0.142) and thus has significantly better predictive probability calibration than both TTM (Brier Score of 0.189) and Logistic Regression (Brier Score of 0.201). The improvements in MedTalk's prediction results can be attributed to MedTalk's use of a Deterministic Random Forest for prediction as opposed to LLM-generated Confidence Estimates that are uncalibrated statistically [2], [14].

C. Clinical User Study

User studies have shown a significant improvement in terms of usability and explainability in clinical settings. In comparison to TTM, MedTalk demonstrated a much better usability score with a 94.2% vs 73.1% success rate in completing tasks [20], which corresponds to a 21 point increase. The time spent on completing tasks was reduced from 63.2 seconds to 134.7 seconds as a result of the MedTalk system, demonstrating that EDAQ's method of providing feedback to users through questions can reduce the effort needed to understand model

predictions [19]. The trust scores provided by the users were significantly higher for MedTalk than for TTM. Users rated the MedTalk explanation at an average score of 6.2 out of 7, while they rated the TTM explanation at 4.1 out of 7 [17]. This is expected because LLM-based explanations may be susceptible to hallucinations [17].

D. Ablation Study

Ablation studies confirm the contribution of each MedTalk component. Removing EDAQ drops our comprehension from 94.2% to 81.3%, underscoring that proactive questioning was the main contributor to interpretability gains. Removing the hybrid explanation mechanism has a small detrimental effect on calibration, whilst disabling fairness guardrails resulted in 7.3% of questions referring to demographic proxies in financial testing tasks [5].

IX. RELATED WORK

A. Dialogue Systems for AI Explainability

TalkToModel [14] represents the first approach to query machine learning models using natural language dialog. The authors integrate a T5-based semantic parser with a BART-based response generator. According to Slack et al. (2023), TalkToModel achieves an average parsing accuracy of about 76%. In addition, according to Slack et al. (2023), 82% of the healthcare practitioners who participated in the survey preferred to use TalkToModel's dialog interface to access explanations of their model's output compared to the traditional dashboard-style interfaces that have been used up until now [3], [14].

Limitations Despite the fact that TTM has had such a major influence on the way we think about model interpretation, TTM is still missing one important component, which is probability calibration analysis; this is very important when you are working in a risk-sensitive domain. Because of the fact that TTM uses generative Large Language Models (LLMs), it can behave in ways that are unpredictable. TTM is mainly limited to supervised tabular tasks. Finally, there are no features in TTM that would allow users to be assured that they are meeting all the relevant regulatory requirements [16]–[18].

Other Dialogue-Adjacent Tools: The Language Interpretability Tool (LIT) [13] provides visualization and interpretability capabilities for NLP models, while the What-If Tool [12] allows for the execution of counterfactual analyses. Neither tool offers the ability to support interactive multi-turn dialogue or deployment frameworks for regulated industries [19], [20].

B. Explainable AI in High-Stakes Domains

Healthcare: Caruana et al. (2015) [3] have been at the forefront of developing interpretable ML technology that will assist in making medical decisions based upon a patient's history and symptoms; particularly with respect to pneumonia risk prediction. Rajkomar et al. (2018) [4] developed extensions of interpretability techniques for use with deep learning models that were trained using Electronic Health

Records (EHRs). Today's literature emphasizes additional factors regarding the development of new ML systems including calibration reliability [5], equitable forecasting [2] and fairness across demographic groups [5].

Finance: Khandani et al. [13] conducted early work illustrating the utility of ensemble methods in evaluating credit risk, and today's U.S. Securities and Exchange Commission (SEC) regulations require that decision-making associated with algorithmic trading be transparent and auditable.

Barocas and Self (2016) [6] analyzed the systemic equity issues embedded within ML-based lending frameworks.

Judiciary: Dressel and Farid (2018) [14] identified significant bias in widely used risk assessment tools such as COMPAS and argued that there should be alternative tools that are transparent and verifiable. Corbett-Davies et al. (2016) [10] also illustrated how fairness and accuracy objectives often conflict when it comes to criminal justice applications.

Recognized Gap: The majority of prior research focuses on domain-specific solutions, resulting in fragmented interpretability efforts. No existing overarching framework provides transparent, conversation-driven explanations that dynamically adapt to the safety constraints and regulatory requirements of multiple sectors simultaneously [5], [19], [20].

X. CONCLUSIONS

MedTalk demonstrates that a deterministic orchestration of transparent sub-components offers a more reliable and auditable approach to explainable AI than generative large language models. By proactively quantifying feature importance and uncertainty through the EDAQ framework, the system achieves both high predictive accuracy (89.4%) and substantially improved clinical trust among end users [7], [19]. The four-layer architecture meets regulatory requirements for auditability and reproducibility, advancing the vision of AI systems that engage in collaborative, verifiable reasoning with human stakeholders [1]–[3].

Future work will extend MedTalk to additional regulated domains including insurance and judiciary, evaluate it on larger and more diverse clinical datasets, and investigate mechanisms for real-time explanation updating as new patient data arrive [5], [20].

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APPENDIX 2: DATASET STATISTICAL SUMMARY

A2.1 PIDD Feature Statistics by Class

Feature	Min	Max	Mean All	Mean Pos.	Mean Neg.	Std Dev	Zero Count
Pregnancies	0	17	3.85	4.87	3.30	3.37	0 (0.0%)
Glucose (mg/dL)	0	199	120.89	141.26	109.98	31.97	5 (0.7%)
Blood Pressure (mmHg)	0	122	69.11	70.82	68.18	19.36	35 (4.6%)
Skin Thickness (mm)	0	99	20.54	22.16	19.66	15.95	227 (29.6%)
Insulin (μU/mL)	0	846	79.80	100.34	68.79	115.24	374 (48.7%)
BMI (kg/m ²)	0	67.1	31.99	35.14	30.30	7.88	11 (1.4%)
Diabetes Pedigree Fn.	0.08	2.42	0.472	0.550	0.430	0.331	0 (0.0%)
Age (years)	21	81	33.24	37.07	31.19	11.76	0 (0.0%)
Outcome (target)	0	1	0.349	1.000	0.000	0.477	500 (65.1%)

Table A2.1: PIDD Feature Statistics – Class-Stratified Summary (Pos. = Diabetic, Neg. = Non-Diabetic)

A2.2 Pearson Correlation Matrix (Selected Feature Pairs)

Feature	Glucose	BMI	Age	Insulin	BP	DPF	Outcome
Glucose	1.00	0.22	0.26	0.33	0.19	0.14	0.47
BMI	0.22	1.00	0.04	0.20	0.28	0.14	0.29
Age	0.26	0.04	1.00	0.06	0.24	0.03	0.24
Insulin	0.33	0.20	0.06	1.00	0.09	0.19	0.13
Blood Pressure	0.19	0.28	0.24	0.09	1.00	0.04	0.07
DPF	0.14	0.14	0.03	0.19	0.04	1.00	0.17
Outcome	0.47	0.29	0.24	0.13	0.07	0.17	1.00

Table A2.2: Pearson Correlation Matrix – Selected PIDD Features (no severe multicollinearity $r > 0.7$)

A2.3 Class Imbalance Analysis

The proportion of diabetic and non-diabetic individuals within the data was 65.1% non-diabetic vs. 34.9% diabetic; this constitutes an imbalance ratio of 1.87:1, which is moderately imbalanced. Without correction, the naive majority-class classifier will have an accuracy of 65.1%; this is the critical baseline that indicates the classifier performs worse than the "trivial" solution.

With correction via SMOTE, the number of instances in each class became roughly equivalent at 500 instances per class. This enabled MedTalk to learn to develop class-specific boundary lines. The improvement in the baseline accuracy (from 65.1%, naive vs. 95.45%, MedTalk w/SMOTE), shows both the effects of SMOTE, class conditional imputation and hyperparameter tuning.

APPENDIX 3: STATUS OF THE PAPER

This paper is currently in the process of being submitted to relevant international conferences for peer review and publication.